Markovian Resolution Limit

What access in time reveals about quantum noise

Explanatory report

A quantum device can appear to lose information irreversibly even when the device and its surroundings evolve together by reversible laws. Information can move into surroundings that we cannot observe. A memoryless noise model describes the visible device without keeping that hidden information. This can be an excellent description of an experiment in which we prepare the device, leave it alone, and read it at the end. But what happens if we touch the device while its interaction with the surroundings is still running?

This report answers a quantitative version of that question. We compare a specified memoryless noise law with microscopic models whose interaction strength is bounded. An experimenter can make a limited number of instantaneous interventions and retain everything the measurements reveal. How many interventions are necessary before some difference becomes observable?

The answer is quadratic in a ratio of physical rates. If the memoryless loss rate is ν and the microscopic strength bound is Λ, then a number of interventions proportional to $(\Lambda/\nu)^2$ is both sufficient and necessary, in the precise worst-case sense developed below, for a fixed distinguishing advantage during one lifetime $1/\nu$. We prove both directions: one experiment detects a difference against every allowed microscopic model, and one microscopic model remains hard to distinguish for every experiment with too few interventions.

The same distinction has a concrete sensing consequence. For a specified device used to estimate an unknown frequency, the number of contacts placed *inside* its microscopic interactions changes the best attainable error. That result needs its own estimator and its own lower bound. Finally, a known fixed pulse displacement and hidden random pulse jitter can have different global consequences even when their timing errors have similar magnitudes. What matters is both the physical interaction and the timing information available to the controller.

How to read this report. The opening develops the quantum and statistical language used in the proofs. The survival experiment supplies the lower bound. The exact collision model supplies the converse. The later sections explain the frequency-estimation application and its timing limitations. The formulas retain the assumptions and resource counts; the surrounding text explains what each formula buys us. No previous reading of the companion sensing report is needed.

There are three questions to keep separate. First, can an experiment tell two descriptions of noise apart? Second, how accurately can a specified device measure an unknown frequency? Third, does the answer change when the controller knows a pulse's timing error, or when that error is hidden? The first asks for a reliable yes-or-no distinction; the second and third ask for a numerical estimate. Better performance on one question does not automatically answer another.

On a first reading, follow the physical question and the explanation around each calculation. The displayed derivations then show why the answer holds, including the difficult cases. A useful recurring test is: what can the experimenter touch, what can they observe, and what must they pay?

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1 States, noise and experiments in time

1.1 A small quantum dictionary, with a physical example

Imagine a two-level device that can be in a ground state $|0\rangle$ or an excited state $|1\rangle$. Such a device is a *qubit*. A pure state is a unit vector, for example

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{1.1}$$

The complex numbers α, β are probability amplitudes. Measuring whether the device is excited gives probability $|\beta|^2$. The angle between the two complex amplitudes, called their relative phase, affects measurements in other bases. It is this phase that later acts as a clock hand. The symbol i satisfies $i^2 = -1$; a dagger denotes conjugate transpose. In column notation $|0\rangle = (1, 0)^{\mathsf{T}}$ and $|1\rangle = (0, 1)^{\mathsf{T}}$, where T means transpose. Complex conjugation replaces $a + ib$ by $a - ib$ for real a, b; conjugate transpose performs that replacement and exchanges rows and columns.

The ket $|\psi\rangle$ is a column vector and the bra $\langle\psi| = |\psi\rangle^\dagger$ is its conjugate row. Their product $\langle\varphi|\psi\rangle$ is an inner product, which measures the overlap of two states; orthogonal states have zero overlap.

A Hermitian matrix satisfies $A^\dagger = A$ and has real eigenvalues. A unitary matrix obeys $U^\dagger U = I$, where I is the identity, and preserves inner products. These are the matrices used for observables and reversible state changes, respectively.

A density matrix ρ represents a state when we may also have classical uncertainty or correlations with an unobserved system. It is positive and has trace one. Positivity means that every quadratic form $\langle v|\rho|v\rangle$ is nonnegative; the trace, written Tr, is the sum of diagonal entries. A pure state has density matrix $|\psi\rangle\langle\psi|$. A probability mixture has $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$, with $p_j \geq 0$ and $\sum_j p_j = 1$. An expectation is written $\mathbb{E}[\cdot]$; it means the average weighted by the actual outcome probabilities.

Some mathematical words will recur. An eigenvector of a matrix A is a nonzero vector v satisfying $Av = \lambda v$; the number λ is its eigenvalue. A subspace is a collection of vectors closed under addition and scalar multiplication. A projector P keeps the component in a chosen subspace and removes the rest, so $P^2 = P = P^\dagger$. The projector $|1\rangle\langle 1|$, for example, keeps the excited component of a qubit.

Three useful Hermitian qubit matrices, called Pauli matrices, are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.2}$$

Thus X exchanges the basis states, while Z changes the sign of $|1\rangle$. Each has measurement outcomes $+1, -1$, whose mean in state ρ is $\mathrm{Tr}(\rho X)$, $\mathrm{Tr}(\rho Y)$ or $\mathrm{Tr}(\rho Z)$, respectively.

Here is a concrete distinction between coherence and classical uncertainty:

$$|+\rangle\langle +| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad |+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \tag{1.3}$$

Both states give equal probabilities for ground and excited outcomes. Their mean X readings differ: one for the first state and zero for the second. The first state's off-diagonal entries carry phase information, called *coherence*; the second is a classical mixture. Equal probabilities in one measurement do not specify a quantum state.

Two systems use a tensor product, written $\otimes$. A joint state need not be a product of individual states. For example,

$$|\Psi\rangle_{SR} = \frac{|0\rangle_S|0\rangle_R + |1\rangle_S|1\rangle_R}{\sqrt{2}} \tag{1.4}$$

is entangled: neither subsystem has its own pure state. The state accessible on S alone is obtained by a partial trace over R, written $\mathrm{Tr}_R(|\Psi\rangle\langle\Psi|) = I_S/2$. The partial trace reproduces every probability for measurements that ignore the traced system. It is not a physical instruction to destroy that system.

For a yes-or-no projective measurement, P is the projector onto the yes subspace. The probability of yes is $\mathrm{Tr}(P\rho)$; after yes, the normalized state is $P\rho P/\mathrm{Tr}(P\rho)$. This is the probability rule and state update we use in the survival experiment. Keeping the result is essential. A conditional statement about successful results is not a statement that unsuccessful runs cost nothing.

1.2 Reversible joint motion and reduced channels

A Hamiltonian H generates reversible evolution $U(t) = e^{-itH}$ when it is time independent. We set $\hbar = 1$, so H has units of inverse time. The state evolves as $\rho \mapsto U\rho U^\dagger$. The exponential is defined by its convergent matrix power series. With a time-dependent Hamiltonian, the propagator instead solves $\mathrm{d}U(t)/\mathrm{d}t = -iH(t)U(t)$ with $U(0) = I$; the ordering of successive interactions matters when the Hamiltonians do not commute.

For a simple clock, let $H = (\epsilon/2)Z$ and initially prepare $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. After time t the state is

$$|\psi(t)\rangle = \frac{e^{-i\epsilon t/2}|0\rangle + e^{i\epsilon t/2}|1\rangle}{\sqrt{2}}, \qquad \langle X\rangle = \cos(\epsilon t), \quad \langle Y\rangle = \sin(\epsilon t). \tag{1.5}$$

The frequency ϵ has units of inverse time and the relative phase ϵt is dimensionless. The two measurement axes read the cosine and sine of that phase. A long duration magnifies a small frequency change, but also permits many whole phase turns; that ambiguity will matter when the frequency is initially unknown.

The visible system is S, the inaccessible environment is E, and the experimenter's protected reference and memory are R. A reduced channel describes the visible input-output rule when the bath starts independently in a fixed state η_E:

$$\mathcal{C}(\rho_S) = \mathrm{Tr}_E\left[U_{SE}(\rho_S \otimes \eta_E)U_{SE}^\dagger\right]. \tag{1.6}$$

A calligraphic symbol such as $\mathcal{C}$ is a linear map on operators, rather than an ordinary matrix acting on state vectors. The channel is completely positive and trace preserving, abbreviated CPTP. Trace preservation keeps total probability one. Complete positivity means that the same rule remains physically valid when S is entangled with any untouched reference.

Equivalently, a channel has a Kraus representation

$$\mathcal{C}(\rho) = \sum_j A_j \rho A_j^\dagger, \qquad \sum_j A_j^\dagger A_j = I. \tag{1.7}$$

For a pure initial bath, $A_j = \langle j|U|0\rangle_E$ is the system operation associated with one bath basis state. This is a mathematical representation; the experimenter does not automatically receive the label j. If distinct labels remain unobserved, their contributions must be summed. Treating them as available measurement results would supply an extra resource.

A quantum instrument is a measurement together with its conditional state update. If the maps for its outcomes x are $\mathcal{A}_x$, then $\mathrm{Tr}\,\mathcal{A}_x(\rho)$ is the probability of x and $\sum_x \mathcal{A}_x$ is CPTP. We retain the classical record explicitly:

$$\rho \longmapsto \sum_x |x\rangle\langle x| \otimes \mathcal{A}_x(\rho). \tag{1.8}$$

The separate terms are subnormalized: their traces already contain the branch probabilities. This form lets later controls depend on earlier results while retaining rare and failed outcomes correctly.

1.3 What a memoryless law asserts

A time-independent Markov generator $\mathcal{L}$ prescribes

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \mathcal{L}(\rho), \qquad \rho(t) = e^{t\mathcal{L}}\rho(0), \qquad e^{(t+s)\mathcal{L}} = e^{t\mathcal{L}}e^{s\mathcal{L}}. \tag{1.9}$$

Here the exponential acts on the space of matrices. Its semigroup property means that elapsed intervals compose by adding their durations. In the *controlled* Markov model studied here, the same channel $e^{(t-s)\mathcal{L}}$ acts between any consecutive contacts at times s, t. Earlier bath information is not another input to that continuation. The experimenter's own memory can still be retained; memoryless noise does not forbid a controller from remembering its outcomes.

For example, amplitude damping describes an excited qubit decaying to its ground state at rate $\gamma > 0$:

$$\mathcal{L}_\gamma(\rho) = \gamma\left(\sigma_-\rho\sigma_+ - \frac{1}{2}\{P,\rho\}\right), \quad \sigma_- = |0\rangle\langle 1|, \quad \sigma_+ = \sigma_-^\dagger, \quad P = |1\rangle\langle 1|. \tag{1.10}$$

The anticommutator is $\{A,B\} = AB+BA$; the commutator is $[A,B] = AB-BA$. Two operators commute when $AB = BA$, and anticommute when $AB = -BA$. An initially excited qubit has survival probability $e^{-\gamma t}$. Its initial loss is therefore γt to first order in time. The word "survival" refers to the outcome of a specified measurement, not to the existence of the particle or device.

An environment that starts independently need not remain independent. Joint evolution builds correlations, and an intermediate intervention can change how those correlations influence the future. Thus matching a final channel, even for every input and every protected reference, does not automatically match experiments that intervene earlier. A *process* specifies those multitime experimental probabilities as well as a final input-output map. This distinction is the object of the report.

1.4 Distances mean differences in observable probabilities

For a matrix A, $\|A\|_\infty$ is its operator norm, the largest factor by which it can stretch a unit vector. For a Hermitian matrix it is the largest absolute eigenvalue. We sometimes omit the subscript when this norm is unambiguous. The trace norm is $\|A\|_1 = \operatorname{Tr}\sqrt{A^\dagger A}$, the sum of singular values. For two states,

$$d(\rho,\sigma) = \frac{1}{2}\|\rho-\sigma\|_1 = \max_{0\le F\le I} |\operatorname{Tr}[F(\rho-\sigma)]|. \tag{1.11}$$

An operator F between zero and the identity represents one outcome of a general measurement. The equality says that trace distance is the largest possible difference in the probability of that outcome. CPTP maps cannot increase this distance: processing two possible states in the same physical way cannot make them more distinguishable than they were before processing.

For classical probability distributions, the corresponding distance is total variation,

$$\mathrm{TV}(p,q) = \frac{1}{2}\sum_x |p(x)-q(x)| = \max_B |p(B)-q(B)|. \tag{1.12}$$

The event B can combine many detailed outcomes. A complete measurement record can therefore be converted into one decision bit without losing the best distinguishability available from that record.

The diamond norm extends a map comparison to arbitrary reference-assisted inputs. For a difference of channels $\mathcal{C}-\mathcal{D}$, $\frac{1}{2}\|\mathcal{C}-\mathcal{D}\|_\diamond$ is the largest output trace distance obtainable by preparing S jointly with a reference and applying the channel only to S. The reference dimension

need not exceed that of the channel input for this optimization. The same norm on linear maps is used for generators and error bounds below. Its key properties here are submultiplicativity under composition and norm one for a CPTP map. These facts justify adding errors without multiplying them by the number of measurement branches.

2 The comparison and the resolution law

2.1 What is fixed, and what is counted

The target Markov law and its calibrated loss rate are given. A competing microscopic device may have an arbitrarily large but finite bath. Its initial state is $\rho_{SR} \otimes \eta_E$: the bath is independent of the input we choose. Its predetermined joint Hamiltonian has the form

$$H_{SE}(t) = I_S \otimes H_E(t) + K(t), \qquad \|K(t)\|_\infty \le \Lambda. \tag{2.1}$$

The inequality holds at almost every time, which permits jumps in a piecewise continuous switching schedule. Every term that affects S, including a system-only drive, belongs to K. Only bath-only motion is unrestricted. Removing H_E by its interaction picture conjugates K unitarily and preserves its norm. It also leaves every tester operation unchanged because the tester never touches E. More explicitly, if $U_E(t)$ is the bath's own propagator, the remaining interaction is

$$K_I(t) = (I_S \otimes U_E^\dagger(t))K(t)(I_S \otimes U_E(t)). \tag{2.2}$$

This is the *bath interaction picture*: coordinates that follow the bath's own motion. Multiplication by a unitary preserves vector lengths, so it preserves the operator norm. The coordinate change supplies no extra physical control.

The device is fixed before the tester chooses its protocol. It cannot choose its initial bath or schedule after seeing the tester's input or measurement results. Ordinary interaction and measurement backaction are allowed. There is no new supply of bath systems after time zero. The construction later uses many bath partners, all initialized at time zero, and retains each one after its interaction ends.

One run lasts a fixed time T. At time zero we may prepare any SR state; at time T we may measure SR and all retained records. Between those times we may make at most N instantaneous contacts with S. A pulse, measurement, reset, or swap into protected memory is a contact. An early readout also counts even if its result is merely stored until T. Operations on R alone are free. Several operations at the same instant can be composed into a single instrument.

The next contact and its time may depend on the previous classical record. The budget is hard: every possible history has at most N contacts. There is no continuous uncounted monitoring of S and no coherent superposition of different contact times. Mathematically we first use finite measurement trees and finite protected memories, of arbitrarily large size. We also include classical mixtures and limits whose output event probabilities converge for the target and each competing device without changing the structural budget. Uniform probability bounds pass to those limits.

2.2 Two questions with different orders of choice

Let T denote one complete tester, and let p_D^{T} and p_M^{T} be the probabilities that its final bit is one under a microscopic device D and the Markov target M. Define

$$\Gamma_N = \sup_{\mathsf{T}} \inf_D (p_D^{\mathsf{T}} - p_M^{\mathsf{T}}), \tag{2.3}$$

$$\Delta_N = \inf_D \sup_{\mathsf{T}} |p_D^{\mathsf{T}} - p_M^{\mathsf{T}}|. \tag{2.4}$$

An infimum is the best lower limit achievable by choices in a set; a supremum is the corresponding upper limit. They need not be attained. For Γ_N, we choose one test first and require its signed gap to be positive against every allowed device. The same bit value must indicate the same hypothesis in every comparison. For Δ_N, the microscopic device is chosen first and the tester can be designed with that device in mind. It asks how well the hardest device can hide from its best test.

The elementary relation is $0 \le \Gamma_N \le \Delta_N \le 1$. A constant-output tester gives zero; the signed performance of any fixed test is bounded by a fixed device's best absolute gap. No equality of the two games is assumed. A signed gap b gives success probability $(1 + b)/2$ when the two hypotheses are assigned equal prior probabilities and output one means "microscopic". Thus $b = 0.1$ means success 0.55, not certainty. Repeated independent runs are an additional resource.

2.3 The theorem and its domain

The general target is a finite-dimensional dissipator

$$\mathcal{L}(\rho) = \sum_a \left(M_a \rho M_a^\dagger - \frac{1}{2}\{M_a^\dagger M_a, \rho\} \right), \qquad Q = \sum_a M_a^\dagger M_a. \tag{2.5}$$

The operators M_a have units of inverse square-root time. The theorem requires a representation in which their means have been optimally centered and no coherent Hamiltonian term remains. In this representation,

$$\nu = \|Q\|_\infty = \max_\rho \sum_a \left[\mathrm{Tr}(\rho M_a^\dagger M_a) - |\,\mathrm{Tr}(\rho M_a)|^2 \right] > 0. \tag{2.6}$$

The maximum is over all system density matrices. Section 4 explains the centering, the maximizing probe, and why the absence of coherent drift is a physical assumption. For amplitude damping, $M = \sqrt{\gamma}\,\sigma_-$ and $\nu = \gamma$.

Take $\Lambda \ge \nu$, set the horizon to one intrinsic lifetime $T = 1/\nu$, and define the dimensionless ratio $q = (\nu/\Lambda)^2 \in (0, 1]$. Then, for every integer $N \ge 0$,

$$\boxed{\frac{1}{64}\min\{1, Nq\} \le \Gamma_N \le \Delta_N \le \min\{1, 2\cosh^2(1) Nq\}.} \tag{2.7}$$

Here $\cosh x = (e^x + e^{-x})/2$. The constants are universal and independent of dimension, but are not optimized. For the qubit amplitude-damping example, the lower coefficient can be improved to 5/64.

The notation $\Theta(f)$ means bounded above and below by positive constant multiples of f throughout the stated asymptotic regime. Equation (2.7) says that both game values have order $\min\{1, Nq\}$. If Nq tends to zero, one admissible device makes every tester ineffective. If N is proportional to $1/q$, one universal test has a fixed positive advantage. Saturation at order one does not claim perfect identification. With $N = 0$, both values are exactly zero because one admissible device reproduces the final channel exactly. It does not mean that every microscopic device has the same final channel.

For a sense of the units, if $\Lambda/\nu = 10$, the natural contact scale is 100. This is an order estimate, not an optimized threshold or a laboratory instruction: the constants in the theorem are deliberately conservative. The calibrated target rate and a justified strength bound are inputs. The protocol does not infer Λ from a decay curve.

The proof now has two distinct obligations. The lower bound must survive every allowed bath and every conditional bath state. The upper bound must supply one device that survives every

allowed tester, including adaptive timing and protected quantum memory. Meeting only one obligation would not establish the matching resolution law.

The direction of these bounds can initially sound backwards. A *lower bound on the observable probability gap* shows that enough contacts suffice to reveal a difference. An *upper bound on the gap* for a carefully built device shows that too few contacts cannot guarantee that distinction, even with a clever experiment. This second direction is the converse. Together they establish how many contacts are necessary and sufficient in order, rather than just supplying one successful protocol.

3 A survival experiment that works without knowing the bath

The lower bound begins with a simple question: after a very short wait, is the system still in the state we prepared? A memoryless decay law and a bounded microscopic Hamiltonian give different answers at the first nonzero order in the waiting time. Repeated observations can accumulate that difference into a measurable probability gap. The challenge is to make the argument work for every allowed environment, including one whose state changes after each observation.

3.1 Linear decay versus quadratic escape

For a decaying qubit, prepare the excited state $|1\rangle$. Write $P = |1\rangle\langle 1|$ for the projector onto that state. A measurement of $\{P, I - P\}$ asks only whether the excitation survived. Under amplitude damping with calibrated population-decay rate $\gamma > 0$, the probability of survival over time t is

$$p_{\mathrm{M}}(t) = e^{-\gamma t} = 1 - \gamma t + O(\gamma^2 t^2). \tag{3.1}$$

The symbol $O(\gamma^2 t^2)$ says that the remainder is bounded in magnitude by a constant times $\gamma^2 t^2$ as t tends to zero. The leading failure probability is proportional to t.

For microscopic unitary evolution, the corresponding initial failure probability is at most $\Lambda^2 t^2$. The square comes from the passage from an amplitude to a probability. A Hamiltonian bounded by Λ can create an amplitude outside the initial subspace at speed at most Λ; the probability of being outside is the squared length of that amplitude. This conclusion does not require the bath to be pure, small, or memoryless.

Here is a precise version. Let Π be the projector onto the desired system–reference subspace, tensored with the identity on the bath. The initial joint state is supported on that subspace. Pass to the interaction picture of the bath-only Hamiltonian. This changes the description of the bath motion while leaving Π fixed and preserving the bound $\|K(t)\| \leq \Lambda$ on the remaining Hamiltonian. A mixed initial state can be represented as part of a larger pure state; the additional purifying system is only a mathematical device.

Let $\ell(t)$ be the probability outside Π. Splitting the evolving state vector into its components inside and outside that subspace gives

$$\left|\frac{\mathrm{d}\ell}{\mathrm{d}t}\right| \leq 2\Lambda\sqrt{\ell(t)[1-\ell(t)]}, \qquad \ell(0) = 0. \tag{3.2}$$

Only the Hamiltonian block connecting the two subspaces can change their probabilities. Its norm is at most Λ; the two component lengths are $\sqrt{\ell}$ and $\sqrt{1-\ell}$, which explains the right-hand side. Write $\ell = \sin^2\vartheta$ with $0 \leq \vartheta \leq \pi/2$. The differential inequality bounds the growth of this

angle by $\mathrm{d}\vartheta/\mathrm{d}t \le \Lambda$. Integrating from the initial angle zero, with the same comparison at the endpoints, gives

$$\ell(t) \le \sin^2(\Lambda t), \qquad p_{\mathrm{D}}(t) \ge \cos^2(\Lambda t) \ge 1 - \Lambda^2 t^2, \quad 0 \le \Lambda t \le \frac{\pi}{2}. \tag{3.3}$$

The subscript D denotes any admissible microscopic device. The final inequality uses $|\sin x| \le |x|$.

After a successful projective measurement, the conditional joint state is again supported on Π. The bath may have changed, and its conditional state may depend on the whole preceding record. Nevertheless, (3.3) applies again because it uses only support in the subspace and the same Hamiltonian bound. For gaps $t_1, \ldots, t_k$ satisfying $0 \le \Lambda t_j \le \pi/2$ for every j, the probability of success at every measurement is therefore at least

$$\Pr_{\mathrm{D}}\{\text{all } k \text{ successes}\} \ge \prod_{j=1}^{k} \cos^2(\Lambda t_j). \tag{3.4}$$

This is the multiplication rule for conditional probabilities, followed by a bound on each factor. It is not an independence assumption. No bath reset occurs between measurements.

Balancing the two leading failure probabilities, γt and $\Lambda^2 t^2$, suggests a waiting time of order γ/Λ^2. A lifetime $1/\gamma$ contains order $(\Lambda/\gamma)^2$ such gaps. This identifies the scale to investigate; the following explicit calculation turns that intuition into a uniform decision rule.

3.2 An explicit experiment for a decaying qubit

Set $r = \Lambda/\gamma \ge 1$ and retain the fixed final horizon $T = 1/\gamma$. Given a contact allowance $N \ge 1$, choose

$$k = \min\{N, \lfloor r^2 \rfloor\}, \qquad \tau = \frac{\gamma}{4\Lambda^2} = \frac{1}{4\gamma r^2}. \tag{3.5}$$

The floor $\lfloor r^2 \rfloor$ is the largest integer not exceeding r^2. Prepare $|1\rangle$ at zero and measure $\{P, I - P\}$ at $\tau, 2\tau, \ldots, k\tau$. Store a classical bit equal to one exactly when every measurement succeeded, and read that stored bit at T. The last system contact occurs by $T/4$. Waiting until the prescribed terminal time still counts toward the full run duration, even though no further system observation is needed.

Every measurement, including the last early measurement, is charged as one contact. A failure sets the stored bit to zero. That run is still an outcome of the experiment. We do not keep only successful runs or renormalize their probabilities.

Under any microscopic device, (3.4) and the elementary inequality $(1-x)^k \ge 1 - kx$ for $0 \le x \le 1$ imply

$$p_{\mathrm{D}} \ge \cos^{2k}\left(\frac{1}{4r}\right) \ge \left(1 - \frac{1}{16r^2}\right)^k \ge 1 - \frac{k}{16r^2}. \tag{3.6}$$

Here p_{D} means the probability that the stored bit is one, not the final excited-state population at T. Under the Markov process, each successful measurement restores the excited state, and the next interval has survival probability $e^{-\gamma\tau}$. Thus

$$p_{\mathrm{M}} = e^{-\gamma k \tau} = e^{-k/(4r^2)}. \tag{3.7}$$

Define $y = k/r^2$, so $0 < y \le 1$. The inequality $e^{-z} \le 1 - z + z^2/2$ for $z \ge 0$ now gives

$$p_{\mathrm{D}} - p_{\mathrm{M}} \ge 1 - \frac{y}{16} - e^{-y/4} \ge \frac{3y}{16} - \frac{y^2}{32} \ge \frac{5y}{32}. \tag{3.8}$$

For every real $r^2 \geq 1$, its integer part is at least $r^2/2$. Consequently $y \geq \frac{1}{2}\min\{1, N/r^2\}$, and

$$\Gamma_N \geq \frac{5}{64}\min\left\{1, \frac{N\gamma^2}{\Lambda^2}\right\}. \tag{3.9}$$

The same event has the same sign against every allowed bath: complete survival is more likely for the bounded microscopic device. That common orientation is what makes it a lower bound on Γ_N, the universal signed-witness quantity.

A numerical scale, without a simulation. Take $\gamma = 1\,\mathrm{s}^{-1}$, $\Lambda = 10\,\mathrm{s}^{-1}$ and $N = 100$. Then $T = 1\,\mathrm{s}$, $k = 100$, and $\tau = 0.0025\,\mathrm{s}$. The last contact is at $0.25\,\mathrm{s}$. Equations (3.6) and (3.7) give

$$p_{\mathrm{D}} \geq 0.9375, \qquad p_{\mathrm{M}} = e^{-1/4} \simeq 0.7788. \tag{3.10}$$

The guaranteed gap is therefore at least $0.9375 - e^{-1/4} \simeq 0.1587$ for this schedule. With equally likely hypotheses, the rule "all successes means microscopic" has success probability at least $[1 + 0.9375 - e^{-1/4}]/2 \simeq 0.5793$. This is an advantage over a coin toss, not certain identification in one run. Estimating these probabilities experimentally requires repeated runs, each with its own duration and contacts.

4 The loss rate and survival witness in a general system

A many-level system may have several dissipative processes, each with a different jump operator. We need one rate that measures their combined initial loss and does not depend on an arbitrary choice of jump labels. We also need a probe that attains that rate. A protected reference is useful here for a concrete reason: the best system state may be mixed, while the survival measurement needs a definite pure joint state.

4.1 From jump fluctuations to an intrinsic rate

For an operator L and a density matrix ρ, define its generalized variance by

$$\mathrm{Var}_\rho(L) = \mathrm{Tr}(\rho L^\dagger L) - |\,\mathrm{Tr}(\rho L)|^2. \tag{4.1}$$

It is nonnegative, even when L is not Hermitian. It measures the part of L's action that cannot be represented by multiplying the state by one complex number. For jumps $L_1, \ldots, L_r$, define

$$\nu = \max_{\rho \in \mathsf{D}(S)} \sum_{a=1}^{r} \mathrm{Var}_\rho(L_a). \tag{4.2}$$

Here $\mathsf{D}(S)$ is the set of system density matrices, and a labels the jump operators. The jumps have units of inverse square root of time; hence ν has units of inverse time. We assume $\nu > 0$.

There is an equivalent way to find ν. Subtract a scalar multiple of the identity from each jump and make the largest remaining squared action as small as possible:

$$\nu = \min_{\alpha \in \mathbb{C}^r} \left\| \sum_a (L_a - \alpha_a I)^\dagger (L_a - \alpha_a I) \right\|. \tag{4.3}$$

The vector α contains r complex numbers; the norm of the positive operator in this expression is its largest eigenvalue. To understand the equality, put

$$F(\rho, \alpha) = \sum_a \mathrm{Tr}\Big[\rho (L_a - \alpha_a I)^\dagger (L_a - \alpha_a I)\Big]. \tag{4.4}$$

At fixed ρ, completing the square gives

$$F(\rho, \alpha) = \sum_a \mathrm{Var}_\rho(L_a) + \sum_a |\alpha_a - \mathrm{Tr}(\rho L_a)|^2. \tag{4.5}$$

Its minimum is attained at $\alpha_a = \mathrm{Tr}(\rho L_a)$. At fixed α, maximizing over density matrices gives the largest eigenvalue in (4.3). The finite-dimensional convex minimax theorem permits interchanging these two optimizations: F is linear in ρ and convex in α, the density matrices form a compact convex set, and the quadratic growth of F confines the minimizing α to a bounded set. This establishes (4.3).

Choose an optimal α and write

$$M_a = L_a - \alpha_a I, \qquad Q = \sum_a M_a^\dagger M_a, \qquad \|Q\| = \nu. \tag{4.6}$$

These are the optimally centered jumps. To see why one state has all the properties we need, choose a variance-maximizing state ρ_* and a norm-minimizing center α_*. The preceding identities give

$$\nu = \min_\alpha F(\rho_*, \alpha) \le F(\rho_*, \alpha_*) = \mathrm{Tr}(\rho_* Q) \le \|Q\| = \nu. \tag{4.7}$$

Every inequality is therefore an equality. In the completed square, each nonnegative squared term must vanish. The expectation of Q can equal its largest eigenvalue only if the state has no weight on smaller eigenvalues. Consequently ρ_* satisfies

$$\mathrm{Tr}(\rho_* M_a) = 0, \qquad \mathrm{Tr}(\rho_* Q) = \nu, \qquad \mathrm{supp}\,\rho_* \subseteq \ker(Q - \nu I). \tag{4.8}$$

The support is the subspace on which ρ_* has nonzero weight; the kernel in the last expression is the top-eigenvalue subspace of Q. The first equality says that the centered jumps have zero mean in this state. The second and third say that the state attains the largest squared action.

4.2 Why a protected reference can be essential

If $\rho_* = \sum_j \lambda_j |j\rangle\langle j|$, introduce a protected reference R_0 and prepare

$$|\Psi_*\rangle = \sum_j \sqrt{\lambda_j}\, |j\rangle_S |j\rangle_{R_0}, \qquad P_* = |\Psi_*\rangle\langle\Psi_*|. \tag{4.9}$$

Tracing out R_0 gives ρ_*. This construction, called a purification, replaces an uncertain system state by a definite joint state whose correlations can be tested. At most $\dim S$ reference levels are required. The reference does not interact with the bath.

For the centered, drift-free generator

$$\mathcal{L}(X) = \sum_a \left(M_a X M_a^\dagger - \tfrac{1}{2}\{M_a^\dagger M_a, X\} \right), \tag{4.10}$$

let $|\phi_a\rangle = (M_a \otimes I)|\Psi_*\rangle$. Equations (4.8) imply $\langle\Psi_*|\phi_a\rangle = 0$ and $(Q \otimes I)|\Psi_*\rangle = \nu|\Psi_*\rangle$. Therefore

$$(\mathcal{L} \otimes \mathrm{id})(P_*) = \sum_a |\phi_a\rangle\langle\phi_a| - \nu P_*. \tag{4.11}$$

The gain term has support orthogonal to the initial joint state and trace ν. The negative term removes weight from that state at rate ν. Thus its survival probability

$$p(t) = \mathrm{Tr}\Big[P_* (e^{t\mathcal{L}} \otimes \mathrm{id})(P_*) \Big] \tag{4.12}$$

satisfies $p(0) = 1$ and $p'(0) = -\nu$, where a prime denotes a time derivative. For a general purification of ρ, the same calculation gives initial loss $\sum_a \mathrm{Var}_\rho(M_a)$, so (4.2) really is the largest attainable initial loss.

The same state also determines the generator's diamond norm:

$$\|\mathcal{L}\|_\diamond = 2\nu. \tag{4.13}$$

For the upper bound, the completely positive gain map $X \mapsto \sum_a M_a X M_a^\dagger$ has diamond norm $\|Q\| = \nu$. Each multiplication map in the anticommutator has norm at most $\|Q\|$, so their half-sum contributes at most another ν. For the reverse bound, the two terms in (4.11) have orthogonal support and traces ν and $-\nu$. Their trace norm is 2ν. This is an allowed reference-assisted input to the diamond norm, establishing equality.

Three calibrations of the same definition. For amplitude damping, $M = \sqrt{\gamma}\,|0\rangle\langle 1|$ and $Q = \gamma|1\rangle\langle 1|$. The excited state attains $\nu = \gamma$. For dephasing with generator $\mathcal{L}(X) = (\kappa/2)(ZXZ - X)$, the off-diagonal density-matrix entries decay as $e^{-\kappa t}$. The jump $M = \sqrt{\kappa/2}\,Z$ gives $\mathrm{Var}_\rho(M) = (\kappa/2)[1 - \mathrm{Tr}(\rho Z)^2]$, so $\nu = \kappa/2$. The state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ attains it.

For depolarization, $\mathcal{L}(X) = \kappa[(I/2)\,\mathrm{Tr}\,X - X]$, take $M_a = \sqrt{\kappa/4}\,\sigma_a$, with $\sigma_x = X$, $\sigma_y = Y$, and $\sigma_z = Z$ the three Pauli matrices. Write a qubit state as $\rho = (I + \mathbf{r}\cdot\boldsymbol{\sigma})/2$, where the real Bloch vector $\mathbf{r}$ has length at most one. Then

$$\sum_a \mathrm{Var}_\rho(M_a) = \frac{\kappa}{4}(3 - |\mathbf{r}|^2). \tag{4.14}$$

The maximum $3\kappa/4$ occurs at $\rho_* = I/2$ and is attained in the survival experiment by a Bell pair $(|00\rangle + |11\rangle)/\sqrt{2}$. A pure system state has $|\mathbf{r}| = 1$ and reaches only $\kappa/2$. The protected reference therefore improves the available loss slope. The natural horizons $1/\nu$ are respectively $1/\gamma$, $2/\kappa$, and $4/(3\kappa)$; one should not identify a named decay parameter with ν without checking its convention.

4.3 A restriction that centering cannot remove

Centering the jumps is a mathematical optimization. It does not erase a physical coherent drive. For a jump $L = M + \alpha I$, write $D[L](X) = LXL^\dagger - \{L^\dagger L, X\}/2$. Expanding the expression gives, for a collection of jumps,

$$\sum_a D[L_a](X) = \sum_a D[M_a](X) - i[H_c, X], \qquad H_c = \frac{i}{2}\sum_a(\alpha_a^* M_a - \alpha_a M_a^\dagger). \tag{4.15}$$

The star denotes complex conjugation. The commutator term is coherent motion that must still be present in the physical generator.

For example, choose

$$L = \sqrt{\gamma}\,(|0\rangle\langle 1| + aI), \qquad a \in \mathbb{R}\setminus\{0\}. \tag{4.16}$$

Its optimal centering gives $M = \sqrt{\gamma}\,|0\rangle\langle 1|$ and $\nu = \gamma$, but leaves

$$H_c = \frac{i\gamma a}{2}(|0\rangle\langle 1| - |1\rangle\langle 0|) = -\frac{\gamma a}{2}Y. \tag{4.17}$$

Increasing $|a|$ makes this coherent motion arbitrarily fast without changing ν. A generator written as one dissipator is therefore not automatically in the centered, drift-free class. The theorem assumes that the coherent term is absent in the centered representation, or that it has been physically compensated as part of the stated model. Changing coordinates cannot supply that compensation for free. In particular, a rapidly rotating frame also rotates the jumps and the survival measurement; the present theorem does not claim uniformity over arbitrary uncharged coherent drift.

4.4 The general lower bound with its constants

Set $q = (\nu/\Lambda)^2 \in (0, 1]$ and $T = 1/\nu$. For $N \geq 1$, use

$$k = \min\{N, \lfloor 1/q \rfloor\}, \qquad \tau = \frac{q}{8\nu} = \frac{\nu}{8\Lambda^2}. \tag{4.18}$$

Prepare P_* and measure $\{P_*, I - P_*\}$ at the k times $\tau, 2\tau, \ldots, k\tau$. Report the all-success bit at T. Every contact is a joint system–reference measurement; its time is strictly interior because $k\tau \leq T/8$.

After a success, the joint system–reference state is the rank-one state P_*. A larger state with this pure marginal must factor as $P_* \otimes \omega_E$. The conditional bath state ω_E can depend on the previous record, but the survival lemma bounds the next microscopic failure probability by

$$\Lambda^2\tau^2 = \frac{q}{64}. \tag{4.19}$$

For the Markov process we need a finite-time remainder bound, not only the initial slope. Twice integrating the derivative of the exponential gives the exact identity

$$e^{t\mathcal{L}} - \mathrm{id} - t\mathcal{L} = \int_0^t (t - s)\mathcal{L}^2 e^{s\mathcal{L}}\, \mathrm{d}s. \tag{4.20}$$

The channel $e^{s\mathcal{L}}$ has diamond norm one, and (4.13) bounds the remaining factors. Hence

$$\|e^{t\mathcal{L}} - \mathrm{id} - t\mathcal{L}\|_\diamond \leq 2\nu^2 t^2, \qquad p(t) \leq 1 - \nu t + 2\nu^2 t^2. \tag{4.21}$$

At the chosen gap,

$$p(\tau) \leq 1 - \frac{q}{8} + \frac{q^2}{32} \leq 1 - \frac{q}{16}. \tag{4.22}$$

The last inequality uses $0 < q \leq 1$. Conditional survival probabilities under the microscopic device can vary, while those under the target repeat the same $p(\tau)$. Both statements suffice to give

$$p_{\mathrm{D}} \geq (1 - q/64)^k, \qquad p_{\mathrm{M}} \leq (1 - q/16)^k. \tag{4.23}$$

For $A \geq B \geq 0$, the identity $A^k - B^k = (A - B)\sum_{j=0}^{k-1} A^{k-1-j}B^j$ is at least $k(A - B)B^{k-1}$. Apply it here and use $kq \leq 1$:

$$\begin{aligned} p_{\mathrm{D}} - p_{\mathrm{M}} &\geq \frac{3kq}{64}(1 - q/16)^{k-1} \geq \frac{45kq}{1024} \\ &\geq \frac{1}{64}\min\{1, Nq\}. \end{aligned} \tag{4.24}$$

The second inequality uses $(1 - q/16)^{k-1} \geq 1 - (k - 1)q/16 \geq 15/16$. For the last one, $kq \geq \frac{1}{2}\min\{1, Nq\}$ and $45/2048 > 1/64$. This proves the universal lower bound with dimension-independent constant $1/64$. If $Nq \geq 1$, the experiment uses only $\lfloor 1/q \rfloor$ contacts and already guarantees a fixed positive gap. The saturation at this scale means an order-one advantage, not perfect discrimination.

4.5 Timing tolerance and imperfect instruments

The survival witness tolerates an interval of waiting times. Suppose each actual gap, possibly selected from previous records, lies in

$$\frac{\nu}{16\Lambda^2} \leq \tau_j \leq \frac{\nu}{8\Lambda^2}, \qquad j = 1, \ldots, k, \quad k = \min\{N, \lfloor 1/q \rfloor\}. \tag{4.25}$$

The total duration of the measured prefix is at most $T/8$. Microscopic survival is at least $1 - q/64$ in every gap. Put $x = \nu\tau_j \in [q/16, q/8]$. Equation (4.21) gives Markov failure at least $x - 2x^2 = x(1-2x) \geq (q/16)(3/4) \geq q/32$. Consequently

$$\begin{aligned} p_{\mathrm{D}} - p_{\mathrm{M}} &\geq (1 - q/64)^k - (1 - q/32)^k \\ &\geq \frac{kq}{64}(1 - q/32)^{k-1} \geq \frac{31kq}{2048} \geq \frac{1}{256}\min\{1, Nq\}. \end{aligned} \tag{4.26}$$

The factor $31/32$ follows again from the inequality $(1-x)^j \geq 1 - jx$ and $kq \leq 1$. Thus exact equality of all spacings is unnecessary. The interval itself is specified using the calibrated ν and certified bound Λ; arbitrary uncertainty in those parameters is not covered.

Finally, suppose the actual complete instrument at contact j differs from its ideal instrument by at most η_j in half diamond norm, uniformly over inputs and protected references. Complete means that the output includes the outcome label as well as the conditional quantum state. Replace the instruments one at a time in the experiment. For either hypothesis, all later channels and instruments contract trace distance, so the final event probability changes by at most $\sum_j \eta_j$. Subtracting the two probabilities yields

$$\text{implemented signed gap} \geq \text{ideal signed gap} - 2\sum_j \eta_j. \tag{4.27}$$

Preparation and readout errors can be included as additional terms in this sum. The bound continues to hold with a retained microscopic bath: the diamond-norm assumption controls inputs entangled with that bath, and subsequent joint evolution still contracts trace distance.

These guarantees concern instantaneous instruments with specified errors. A finite-duration laboratory pulse must first be shown to implement such an instrument while the bath acts during the pulse. That implementation is an additional problem. The witness establishes what sufficiently accurate contacts can reveal; the next construction establishes why fewer contacts cannot always reveal more.

5 A microscopic device that hides between contacts

The survival experiment proves that sufficiently frequent interventions can reveal bounded microscopic motion. To establish the optimal contact scale, we must also answer the experimenter's strongest objection: could a much cleverer experiment reveal it with far fewer contacts? We answer by building one microscopic device that remains close to the target for *every* allowed low-contact experiment. Its bath, Hamiltonian and switching schedule are fixed before an experiment is chosen. The experimenter may know that schedule and may choose contacts using earlier measurement results.

The construction has two requirements. Without a contact inside a short interaction interval, it must give exactly the target channel. If contacts do occur inside that interval, their complete recorded output must still remain close to the target. Exact agreement at the endpoints alone does not establish the second requirement. We first see both issues in a two-qubit example, then build the general device.

5.1 One exchange reproduces a decay channel exactly

Let the sensor be a qubit and let a bath qubit begin in $|0\rangle_E$. During a cell of duration $h > 0$, use the exchange Hamiltonian

$$K = \frac{\theta}{h}\left(\sigma_+ \otimes \sigma_- + \sigma_- \otimes \sigma_+\right), \qquad \cos\theta = e^{-\gamma h/2}, \qquad 0 < \theta < \frac{\pi}{2}. \tag{5.1}$$

Here $\sigma_- = |0\rangle\langle 1|$ removes an excitation and $\sigma_+ = \sigma_-^\dagger$ adds one. The tensor product places the first operator on the sensor and the second on the bath. The interaction can transfer an excitation between them; it does nothing to $|0\rangle|0\rangle$. Its norm is θ/h, with units of inverse time.

At the end of the cell, the unitary $U = e^{-ihK}$ acts on the two possible sensor basis inputs as

$$\begin{aligned} |0\rangle|0\rangle &\longmapsto |0\rangle|0\rangle, \\ |1\rangle|0\rangle &\longmapsto \cos\theta\,|1\rangle|0\rangle - i\sin\theta\,|0\rangle|1\rangle. \end{aligned} \tag{5.2}$$

The inaccessible bath carries away the distinction between these two output alternatives. Ignoring it gives the sensor channel $\rho \mapsto A_0\rho A_0^\dagger + A_1\rho A_1^\dagger$, where

$$A_0 = |0\rangle\langle 0| + \cos\theta\,|1\rangle\langle 1|, \qquad A_1 = -i\sin\theta\,|0\rangle\langle 1|. \tag{5.3}$$

The excited population is multiplied by $\cos^2\theta = e^{-\gamma h}$, and a coherence between $|0\rangle$ and $|1\rangle$ is multiplied by $\cos\theta = e^{-\gamma h/2}$. These are exactly the amplitude-damping rules for time h. Checking population alone would not suffice: the coherence rule is also needed. The channel identity implies equality on every sensor state, including a sensor entangled with a protected reference.

The joint evolution during the interval remains different. For example, an initially excited sensor has microscopic excited-state probability $\cos^2(\theta t/h)$ at intermediate time t, whereas the target has $e^{-\gamma t}$. The two formulas agree at $t = h$ by construction. They have different initial slopes: zero for the microscopic model and $-\gamma$ for the target. The endpoint equality therefore leaves room for an interior intervention to expose the difference.

An endpoint channel is like a rule for the final reading of an instrument. It does not specify what happens if we open the instrument halfway through its operation. Here the bath still carries the excitation coherently during the interaction, and an interior pulse can change how it returns.

5.2 A single interior pulse exposes that difference

Apply $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$ to the sensor halfway through the cell. This pulse changes the sign of the exchange Hamiltonian: $(Z \otimes I)K(Z \otimes I) = -K$. If $U_{1/2} = e^{-ihK/2}$, then

$$U_{1/2}(Z \otimes I)U_{1/2} = Z \otimes I. \tag{5.4}$$

One way to verify the identity is to move $Z \otimes I$ through the right-hand exponential. Anticommutation changes that exponential to $e^{+ihK/2}$, which cancels the left-hand one. The excitation exchange is undone. Apart from the known Z pulse, the sensor and bath return to their input state.

The same pulse cannot undo Markovian amplitude damping. Changing the sign of σ_- does not change its dissipator, since each noise term contains both σ_- and its adjoint. Thus the amplitude-damping channel commutes with the phase flip, giving

$$e^{(h/2)\mathcal{L}_\gamma} \circ \mathrm{Ad}_Z \circ e^{(h/2)\mathcal{L}_\gamma} = \mathrm{Ad}_Z \circ e^{h\mathcal{L}_\gamma}, \qquad \mathrm{Ad}_Z(\rho) = Z\rho Z. \tag{5.5}$$

For an initially excited sensor, the microscopic output is still excited with probability one. The target output is excited with probability $e^{-\gamma h}$. Measuring the population therefore gives a probability gap, equal here to the output trace distance, of

$$1 - e^{-\gamma h} = \sin^2\theta. \tag{5.6}$$

For example, if $\gamma h = 0.01$, the gap is approximately 0.00995. The example shows why a complete microscopic interaction cannot be replaced by its endpoint channel before inserting the pulse. It also shows a discrepancy quadratic in the small action θ. It does not yet bound all possible instruments or show how much discrepancy many adaptive contacts can accumulate.

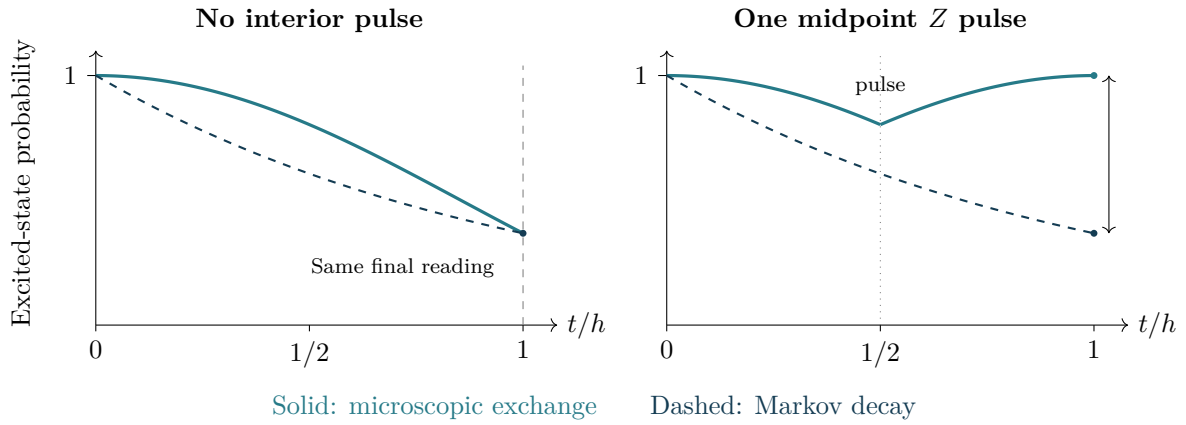


Figure 1: The same endpoint can conceal different dynamics. These are exact excited-population curves for the example $\gamma h = 1$, with $\theta = \arccos(e^{-1/2})$. Without the pulse, both final populations are e^{-1}. With the pulse, the microscopic exchange is undone and its population returns to one; the Markov decay continues. The proof above establishes equality of the whole endpoint channel in the unpulsed experiment, not just equality of these plotted populations.

5.3 An exact cell for a general dissipator

Return to the centered, drift-free generator and its positive loss operator $Q = \sum_a M_a^\dagger M_a$, with $\|Q\| = \nu$. The system dimension is d. We seek a finite bath and a time-independent Hamiltonian for one cell whose reduced endpoint channel is exactly $e^{h\mathcal{L}}$. The same Hamiltonian will also have the symmetry needed to bound interrupted cells.

First isolate one particularly useful Kraus operator,

$$A_0 = e^{-hQ/2}. \tag{5.7}$$

It is a positive operator: all its eigenvalues are positive real numbers. It represents the no-jump contribution. This is an algebraic decomposition of the channel, not a claim that the tester observes environmental jumps.

To justify it, split the generator into

$$\mathcal{L} = \mathcal{G} + \mathcal{J}, \qquad \mathcal{G}(\rho) = -\tfrac{1}{2}\{Q, \rho\}, \qquad \mathcal{J}(\rho) = \sum_a M_a \rho M_a^\dagger. \tag{5.8}$$

The propagation $e^{s\mathcal{G}}(\rho) = e^{-sQ/2}\rho e^{-sQ/2}$ and the map $\mathcal{J}$ are completely positive. Expanding the full propagator according to the number of inserted $\mathcal{J}$ maps yields

$$e^{h\mathcal{L}} = e^{h\mathcal{G}} + \int_0^h e^{(h-t)\mathcal{G}} \mathcal{J} e^{t\mathcal{G}}\, \mathrm{d}t + \text{terms with two or more insertions.} \tag{5.9}$$

Each later term is an ordered integral of completely positive maps. The series converges in finite dimension. Consequently the remainder after subtracting $A_0(\cdot)A_0$ is completely positive. Nothing in this argument assumes that the jumps commute with one another or with Q.

Choose Kraus operators $B_1, \ldots, B_s$ for this remainder, with $s \le d^2$, and regard them as one column operator B. In plain terms, B sends a system vector to the column containing all the vectors B_a would produce. Because the entire channel preserves trace,

$$A_0^2 + B^\dagger B = I. \tag{5.10}$$

This is the condition that the total probability in the no-jump and remaining branches adds to one.

We now convert that column into an actual unitary motion. In a scalar example, if a real amplitude $0 \le a \le 1$ remains in the original sector, the amplitude transferred elsewhere must have length $\sqrt{1-a^2}$. A rotation through $\arccos a$ has precisely these cosine and sine components. We apply that construction in each eigenvector direction of A_0. Define the angle operator $\Theta = \arccos A_0$ by diagonalizing A_0 and applying arccos to each eigenvalue. Its eigenvalues lie in $[0, \pi/2)$. The polar decomposition of B has the form

$$B = J \sin\Theta, \qquad J^\dagger J = P_\Theta, \tag{5.11}$$

where P_Θ projects onto the support of Θ: the subspace with nonzero angle. The map J preserves lengths on that subspace and maps it into the excited bath sectors. It may vanish on the orthogonal subspace; this is why it is called a partial isometry. If an angle is zero, there is no amplitude to rotate into the bath. We never divide by that angle or by its sine, so a rank-deficient loss operator causes no singularity.

Give the bath a vacuum state $|0\rangle$ and s excited labels $|1\rangle, \dots, |s\rangle$. Decompose the joint Hilbert space into the vacuum sector and the direct sum of the excited sectors. On this space set

$$K_h = \frac{1}{h} \begin{pmatrix} 0 & -i\Theta J^\dagger \\ iJ\Theta & 0 \end{pmatrix}. \tag{5.12}$$

The off-diagonal blocks are adjoints of one another, so K_h is a Hermitian Hamiltonian. For each eigenvector with angle $\vartheta > 0$, it generates an ordinary two-dimensional rotation between that vector in the vacuum sector and its image under J. The zero-angle sector and unused excited directions are left fixed. Therefore

$$e^{-ihK_h} \begin{pmatrix} I \\ 0 \end{pmatrix} = \begin{pmatrix} \cos\Theta \\ J\sin\Theta \end{pmatrix} = \begin{pmatrix} A_0 \\ B \end{pmatrix}. \tag{5.13}$$

This equation specifies the unitary's action for a vacuum bath and every system input. Taking the bath trace gives precisely the desired Kraus channel, and hence precisely $e^{h\mathcal{L}}$.

Its norm is controlled by the largest rotation angle:

$$h\|K_h\| = \|\Theta\| = \arccos(e^{-\nu h/2}) =: \theta_h, \qquad \theta_h^2 \le \nu h. \tag{5.14}$$

For the last inequality, use $-2\log\cos\theta = 2\int_0^\theta \tan u \, \mathrm{d}u \ge \theta^2$, because $\tan u \ge u$ for $0 \le u < \pi/2$. Here log is the natural logarithm. This proves $\|K_h\| \le \sqrt{\nu/h}$.

Finally, let bath parity assign sign $+1$ to its vacuum and sign -1 to every excited label:

$$Z_E = |0\rangle\langle 0| - \sum_{a=1}^{s} |a\rangle\langle a|. \tag{5.15}$$

Conjugating K_h by $I_S \otimes Z_E$ changes the signs of both off-diagonal blocks, so it sends K_h to $-K_h$. The initial bath state is unchanged. This symmetry is the extra property that will control arbitrary interruptions; an arbitrary endpoint dilation would not automatically supply it.

5.4 From one cell to one admissible finite bath

Recall $T = 1/\nu$, $\Lambda \ge \nu$ and $q = (\nu/\Lambda)^2$. Choose the number of collision cells and their duration as

$$M_c = \lfloor 1/q \rfloor, \qquad h = \frac{T}{M_c}, \qquad x = \nu h = \frac{1}{M_c}. \tag{5.16}$$

The floor $\lfloor y \rfloor$ denotes the largest integer no larger than y. Since $1/q \ge 1$, we have $M_c \ge 1$ and $x \le 1$. Also $\lfloor y \rfloor \ge y/2$ for $y \ge 1$, so $x \le 2q$.

Prepare all M_c bath cells in their vacuum states at time zero. Couple the system to cell 1 for the first interval, to cell 2 for the second, and so on. Each cell is retained after use but never coupled again. "Fresh" here means previously unused, not newly supplied or reset during the experiment. The strength satisfies

$$\|K_h\| \le \sqrt{\frac{\nu}{h}} = \nu\sqrt{M_c} \le \Lambda. \tag{5.17}$$

The model is therefore admissible. Crucially, neither M_c nor the switching schedule depends on the contact budget N, the chosen input, or a measurement outcome. This one device will supply the upper bound for every N.

An untouched cell has exactly the target endpoint action. Previously used cells may be correlated with the tester, but never interact again; tracing them out cannot change later tester statistics. Future cells remain in their initial independent states. These facts allow us to treat each next cell as a channel on the accessible system, memory and records. They are properties of this construction, not assumptions that an arbitrary physical bath can be reset.

5.5 Why even a complicated interrupted cell has small error

Fix any allowed control subroutine during one cell. Keep its input, protected reference, instruments, classical records and feedback rules identical in three experiments: the microscopic cell, the target Markov cell, and an idle cell in which the system does not evolve between contacts. Denote their complete output channels by $\mathcal{F}_D$, $\mathcal{F}_M$ and $\mathcal{F}_0$. The idle experiment is a mathematical comparison; it is not an extra noise-free setting granted to the tester.

Start with a finite adaptive protocol tree. Collect all the contact times appearing on its branches into one ordered list. At each time apply the prescribed instrument controlled by the earlier record, using the identity on branches that have no contact then. Retain every outcome. This represents the same adaptive experiment as a sequence of trace-preserving channels. It does not charge the union of those times as the contact budget; it is only a convenient representation for the following single-cell estimate.

Multiply the microscopic Hamiltonian by a real scalar α and expand the resulting output channel as a power series,

$$\mathcal{F}_{\alpha K_h} = \sum_{k=0}^{\infty} \alpha^k \mathcal{F}^{(k)}, \qquad \|\mathcal{F}^{(k)}\|_\diamond \le \frac{(2\theta_h)^k}{k!}. \tag{5.18}$$

Each insertion of the Hamiltonian commutator has diamond norm at most $2\|K_h\|$. The maps between insertions have norm one. Integrating over ordered insertion times supplies the factor $h^k/k!$, which proves the displayed estimate. The bound includes arbitrarily large finite protected references and all classical outputs.

Bath parity changes α to $-\alpha$. It leaves the initial vacuum unchanged, commutes with every tester operation, and disappears when the bath is finally traced out. Hence $\mathcal{F}_{\alpha K_h} = \mathcal{F}_{-\alpha K_h}$. Every odd coefficient in the power series must vanish. Since its zero-order term is the idle experiment, summing the remaining terms gives

$$\frac{1}{2}\|\mathcal{F}_D - \mathcal{F}_0\|_\diamond \le \frac{1}{2}\sum_{k=1}^{\infty} \frac{(2\theta_h)^{2k}}{(2k)!} = \sinh^2\theta_h. \tag{5.19}$$

Here $\sinh u = (e^u - e^{-u})/2$ and $\cosh u = (e^u + e^{-u})/2$. The missing linear term matters: θ_h is of order $\sqrt{x}$, whereas the desired discrepancy is of order x.

For the target-to-idle comparison, replace the Markov evolution by idle one infinitesimal interval at a time. The unchanged preceding and following physical maps contract trace distance.

Integrating the generator norm $\|\mathcal{L}\|_\diamond = 2\nu$ over the cell yields

$$\frac{1}{2}\|\mathcal{F}_M - \mathcal{F}_0\|_\diamond \le \nu h = x. \tag{5.20}$$

The triangle inequality now bounds the actual cell discrepancy by

$$\frac{1}{2}\|\mathcal{F}_D - \mathcal{F}_M\|_\diamond \le \sinh^2\theta_h + x \le \cosh^2(1)x =: \varepsilon_{\text{cell}}. \tag{5.21}$$

Indeed $\theta_h \le \sqrt{x} \le 1$, and convexity gives $\sinh u \le u \sinh 1$ for $0 \le u \le 1$. Combining this with $\cosh^2 1 = 1 + \sinh^2 1$ proves the last inequality. The estimate is independent of how many contacts the subroutine makes inside this cell. It concerns the complete output with its failure branches included, never a normalized rare selected outcome.

5.6 Adaptive timing still pays only for the cells actually touched

Call a cell *active* if it contains a system contact strictly inside its interval. A contact exactly at a cell boundary still uses the tester's budget, but lies between two matched endpoint channels. It need not be charged as an active cell. On any realized history, at most N cells are active. The difficulty is that different histories can activate different cells. Their union might contain far more than N cells.

For example, suppose a fair coin chooses whether the sole contact occurs in cell 1 or cell 2. Two cells appear in the list of all possibilities, but each actual run uses one contact. Their activation probabilities are $p_1 = p_2 = 1/2$, and $p_1 + p_2 = 1$. The proof must preserve this accounting even when the choice is made by a measurement on an entangled memory.

We first put the decision to activate a cell at a well-defined place. Before its first system contact, the tester can only act on its protected memory R. A branch map $\mathcal{A}_R$ commutes with any joint system–bath propagation $\mathcal{U}_{SE}$:

$$(\mathcal{U}_{SE} \otimes \mathrm{id}_R)(\mathrm{id}_{SE} \otimes \mathcal{A}_R) = (\mathrm{id}_{SE} \otimes \mathcal{A}_R)(\mathcal{U}_{SE} \otimes \mathrm{id}_R). \tag{5.22}$$

This operator identity holds on entangled inputs too. A memory measurement may change the conditional system–bath state, but moving the same branch map through the propagation preserves that subnormalized state and its probability. Commutation is not an independence assumption.

Collect the memory-only operations up to the first system contact or the cell's end, whichever comes first. Move those operations to the entrance using (5.22), retaining their outcomes and the originally selected first-contact time. The resulting entrance instrument declares either an active cell and its first-contact time, or no contact inside the cell. Nothing from after the cell is moved backward. Later contacts in an active cell remain fully adaptive. This is a representation of the same controller, not an extra information channel or a free move of a physical system contact.

Let Φ_j and Ψ_j be the complete microscopic and target channels for cell j, including that common entrance instrument and the tester's subroutine. For an input state ρ, let $p_j(\rho)$ be its active-flag probability. The inactive outputs agree exactly. The active outputs obey (5.21); multiplying their normalized bound by the probability of that branch gives

$$\frac{1}{2}\|(\Phi_j - \Psi_j)(\rho)\|_1 \le \varepsilon_{\text{cell}} p_j(\rho). \tag{5.23}$$

Complete records keep different entrance outcomes in separate classical blocks, so these weighted estimates add without discarding information.

To compare the whole experiment, replace its cells one at a time using the telescoping identity

$$\Phi_{M_c}\cdots\Phi_1-\Psi_{M_c}\cdots\Psi_1=\sum_{j=1}^{M_c}\Psi_{M_c}\cdots\Psi_{j+1}(\Phi_j-\Psi_j)\Phi_{j-1}\cdots\Phi_1. \tag{5.24}$$

In each summand the preceding part is the genuine microscopic execution, and the following part is a target channel that contracts the discrepancy. Thus every activity weight comes from one and the same probability law, the full microscopic experiment. Writing p_j for those probabilities,

$$\begin{aligned}\frac{1}{2}\|\rho_D^{\mathsf{T}}-\rho_M^{\mathsf{T}}\|_1&\le\varepsilon_{\mathrm{cell}}\sum_{j=1}^{M_c}p_j,\\ \sum_{j=1}^{M_c}p_j&=\mathbb{E}_D[\text{number of active cells}]\le N.\end{aligned} \tag{5.25}$$

The expectation is bounded because every history obeys the hard contact budget. A rare branch does not receive free contacts, and mutually exclusive branches are not counted as though they occurred together.

Finally, $x=1/M_c\le 2q$ and a final binary readout cannot increase trace distance. The constructed device therefore obeys, for every allowed tester,

$$|p_D^{\mathsf{T}}-p_M^{\mathsf{T}}|\le\min\{1,2\cosh^2(1)Nq\}. \tag{5.26}$$

The argument directly covers finite protocol trees at arbitrary real contact times. The bound survives the stipulated operational limits of such controllers, since it is uniform and their output probabilities converge. This does not enlarge the model to coherent superpositions of contact times or uncounted continuous monitoring.

5.7 What the two directions establish

Combining this single-device upper bound with the universal survival witness gives

$$\frac{1}{64}\min\{1,Nq\}\le\Gamma_N\le\Delta_N\le\min\{1,2\cosh^2(1)Nq\},\qquad q=\frac{\nu^2}{\Lambda^2}. \tag{5.27}$$

The lower tester does not know the microscopic bath. The upper simulator does not know which tester will be used. These complementary uniform statements establish a contact cost of order $(\Lambda/\nu)^2$ for a fixed positive single-run distinguishing advantage. They do not assert an exact threshold or optimal numerical constants.

At $N=0$, every cell is inactive, so their exact endpoint channels compose to $e^{T\mathcal{L}}$. This particular microscopic model is indistinguishable from the target by every terminal measurement, including reference-assisted ones. Hence $\Gamma_0=\Delta_0=0$. This is an existence statement about a hiding model; other microscopic models may already have a different endpoint. An earlier system readout is a paid interior contact, even if its result is stored until T.

The same theorem can be read as a strength requirement for simulation. Let $\Lambda_{\min}(N,\epsilon)$ be the infimum of allowed strengths for which some microscopic model keeps the worst permitted tester's probability gap at most ϵ. For $N\ge 1$ and $0<\epsilon<1/64$, the lower bound forces its unsaturated branch; otherwise its gap would already be at least $1/64$. Inverting the two inequalities gives

$$\nu\sqrt{\frac{N}{64\epsilon}}\le\Lambda_{\min}(N,\epsilon)\le\nu\sqrt{\frac{2\cosh^2(1)N}{\epsilon}}. \tag{5.28}$$

In words, tolerating more interventions or demanding smaller error requires a larger microscopic strength, proportional to $\nu\sqrt{N/\epsilon}$ in this model.

There are costs outside that statement. Each general cell has at most d^2+1 bath levels, so the complete bath used here has dimension at most $(d^2+1)^{M_c}$. For the explicit qubit exchange construction it is 2^{M_c}. All those cells are prepared at time zero; the schedule is externally prescribed. The result neither bounds the effort of preparing that bath nor supplies an autonomous clock or finite-duration instruments. Independent repetitions of the whole duration-T experiment are also additional resources. The theorem is a precise statement about interaction strength and counted contacts, with these other resources stated openly.

We have now completed the discrimination argument. The next question asks what temporal access can *enable* in an estimation task. A visible difference between two processes does not itself provide a precise frequency estimate. That requires a separate device, estimator and lower-bound argument.

6 What a record comparison does, and does not, tell us about sensing

The resolution theorem compares two descriptions of a controlled experiment. An estimation theorem asks how accurately a number can be inferred from its data. These questions are related, but neither answers the other by itself. A measurement can reveal that a noise model is inaccurate without revealing much about the frequency we want. Conversely, microscopic memory can make the actual sensor more useful than the memoryless model predicts.

This section makes the connection precise for one simple device. It also explains why the global sensing theorem that follows needs an explicit estimator, a fixed frequency band and a separate lower bound.

6.1 A parity record can test the noise model

Take a sensor qubit S and a protected ancilla qubit A; an ancilla is an additional quantum system used to prepare or read the experiment. Use the Pauli matrices defined in (1.2). The operator Z_SZ_A means $Z_S\otimes Z_A$. Its even sector is spanned by $|00\rangle, |11\rangle$, and its odd sector by $|10\rangle, |01\rangle$. Measuring its sign is a parity measurement. It tells us whether a transverse sensor flip occurred an odd number of times, without distinguishing the two basis states inside the same sector. The even projector is $\Pi = (I + Z_SZ_A)/2$.

A transverse error has the form

$$N_\phi = \cos\phi\, X_S + \sin\phi\, Y_S, \qquad N_\phi^2 = I, \qquad N_\phi Z_S N_\phi = -Z_S. \tag{6.1}$$

The angle ϕ is unknown but fixed across the experiment. Two identical flips cancel, while a flip reverses the sign of subsequent signal phase. Let $\epsilon = \omega - \omega_0$ be the unknown frequency relative to a fixed nominal compensation ω_0. The compensation is a physical setting; calculating a new classical estimate does not change it.

For this bridge, a Markov device accepts a classical rate command $\gamma \in \{a, b\}$, with fixed known $0 < a < b$, and has generator

$$\mathcal{L}_{\epsilon,\phi,\gamma}(\rho) = -i\frac{\epsilon}{2}[Z_S, \rho] + \gamma(N_\phi \rho N_\phi - \rho). \tag{6.2}$$

The rate command is an explicit input port of this model. It is not an extra free control secretly inserted into the preceding fixed-generator theorem. The sensing theorem in the next section will instead hold one microscopic coupling fixed.

Start even, check parity every Δ units of time, and use rate a after an even result and b after an odd result. Let E_n be the event that all n checks are even, with $T = n\Delta$. This event includes hidden pairs of errors between checks; it is not a no-error event. Under the Markov law, every interval along the all-even history uses rate a. With that setting, the instantaneous jump rate is a, independent of the sensor state, because $N_\phi^\dagger N_\phi = I$. The number K of jumps in one interval has the Poisson probabilities

$$\Pr(K = k) = e^{-a\Delta}\frac{(a\Delta)^k}{k!}, \qquad k = 0, 1, 2, \ldots. \tag{6.3}$$

Here $k! = 1 \cdot 2 \cdots k$, with $0! = 1$, and the mean count is $a\Delta$. Summing the even terms gives

$$\Pr_M(E_n) = \left(\frac{1 + e^{-2a\Delta}}{2}\right)^n. \tag{6.4}$$

For example, the identity follows by adding the exponential series for $e^{a\Delta}$ and $e^{-a\Delta}$ and dividing by two, then multiplying by $e^{-a\Delta}$. The signal preserves parity and cannot change this probability.

Now suppose a competing microscopic device has a parity-preserving Hamiltonian part and a parity-changing part bounded by $\Lambda_\perp$. Bath-only motion and the signal can be included in the preserving part. The same subspace-survival argument as before gives

$$\Pr_D(E_n) \geq \cos^{2n}(\Lambda_\perp \Delta), \qquad 0 \leq \Lambda_\perp \Delta \leq \pi/2. \tag{6.5}$$

It remains valid after each success despite the conditional bath state. Consequently the total variation of the *complete* record distributions is at least

$$\max\left\{0, \cos^{2n}(\Lambda_\perp \Delta) - \left(\frac{1 + e^{-2a\Delta}}{2}\right)^n\right\}. \tag{6.6}$$

At fixed T and fixed $\Lambda_\perp$, taking $\Delta = T/n \to 0$ makes this certificate tend to $1 - e^{-aT}$. At fixed positive Δ and growing T, both all-even probabilities may instead tend to zero. The order of limits therefore matters. Also, a bound on parity-changing noise is not the complete system-affecting bound Λ of (2.1); it need not constrain a large diagonal signal.

6.2 One microscopic device approximating complete records

For a specified cell duration h, choose the coupling associated with a rate command as

$$p_\gamma = \frac{1 - e^{-2\gamma h}}{2}, \qquad \theta_\gamma = \arcsin\sqrt{p_\gamma}, \qquad g_\gamma = \frac{\theta_\gamma}{h}. \tag{6.7}$$

Each cell couples the sensor to one bath qubit initially in $|0\rangle$, using

$$H_\gamma = \frac{\epsilon}{2} Z_S \otimes I_E + g_\gamma N_\phi \otimes X_E. \tag{6.8}$$

All bath qubits are supplied at time zero and retained after use. At $\epsilon = 0$, tracing the bath gives exactly the Markov cell channel in (6.2): its flip probability is p_γ. For $|\epsilon| \leq W$, the complete interaction norm is at most $\sqrt{g_b^2 + W^2/4}$. Reporting g_b alone as that norm would omit the signal.

With the signal present, the exact collision channel $\mathcal{C}_\gamma$ and the Markov channel obey the uniform one-cell estimate

$$\frac{1}{2}\|\mathcal{C}_\gamma - e^{h\mathcal{L}_{\epsilon,\phi,\gamma}}\|_\diamond \leq \frac{|\epsilon|h}{2}[\theta_\gamma \sinh(2\theta_\gamma) + \gamma h] \leq 2|\epsilon|\gamma h^2. \tag{6.9}$$

Here $\sinh x = (e^x - e^{-x})/2$. To see the important small-time order, remove the signal by its interaction picture. The rotating error operator satisfies $\|N_\phi(t) - N_\phi(0)\| \le |\epsilon| t$. Odd bath terms vanish by the same parity symmetry used for the controlled-cell theorem. At even order k of the time-ordered expansion, telescoping the k coupling factors and integrating their time arguments bounds the half-diamond difference by terms whose sum is

$$\frac{|\epsilon| h}{4} \sum_{\substack{k \ge 2 \\ k \text{ even}}} \frac{k (2\theta_\gamma)^k}{k!} = \frac{|\epsilon| h \theta_\gamma}{2} \sinh(2\theta_\gamma). \tag{6.10}$$

The rotating Markov dissipator contributes at most $\gamma |\epsilon| h^2/2$ by the contractive integral comparison. Finally, $\theta_\gamma < \pi/4$ and $\theta_\gamma^2 \le \gamma h$ give the last bound in (6.9). The cancellation of odd terms is what avoids a weaker square-root-in-time estimate.

Restrict commands and system instruments to cell boundaries. Keeping all records and replacing cells one at a time then gives, over T/h cells,

$$\mathrm{TV}(P_D, P_M) \le \delta(T, h), \qquad \delta(T, h) = \min\{1, 2WbTh\}. \tag{6.11}$$

Each complete adaptive boundary instrument is a channel and contracts the error. The uniform cell bound applies to every conditional input and selected rate, so no number-of-branches factor appears. Readouts finishing early can be stored through the remaining charged horizon. Inside-cell operations are outside this particular comparison.

A *loss* is a numerical penalty assigned to an estimate; its average is the *risk*. Squared frequency error is the loss used below. If a loss takes values between zero and B, its expectation changes by at most $B\,\delta(T, h)$ between these two record laws. For a frequency interval of width L, clipping the estimate to the interval makes squared loss at most L^2. Thus the *same* estimator and experiment have risks differing by at most $L^2 \delta(T, h)$. The same conclusion survives optimization if both models use the same strategy class and parameter domain. Clipping means replacing an answer outside the interval by its nearest endpoint. This cannot move the answer farther from a true frequency inside the interval.

This is a finite-horizon error-transfer bound. Two models can predict almost the same records while their remaining difference is still larger than the tiny estimation error we seek. The comparison therefore cannot replace a direct precision proof. It also does not say that a small total-variation error controls frequency derivatives. For instance, a small function can vary rapidly as its argument changes. Local Fisher information depends on derivatives of probabilities and therefore needs another argument. Likewise, taking h smaller improves (6.11) but increases the required coupling, $g_\gamma \sim \sqrt{\gamma/h}$, as well as the number of preinitialized bath cells. A fixed-strength global sensing result cannot be obtained by making these resources grow without charging them. We now hold the hardware fixed and solve the estimation task directly.

6.3 Why an identifiable frequency band is a real condition

Watching a clock hand only at fixed intervals can hide whole extra turns. Different rotation speeds can then produce exactly the same readings. In estimation this is called an *alias*. An identifiable band is a promised frequency interval on which the available observations can distinguish different frequencies. More data cannot resolve an exact alias if every new observation repeats the same ambiguity.

Fix one coupling $g > 0$ and cell duration h, let $\chi = gh < \pi/4$, and retain the simultaneous Hamiltonian $H = (\epsilon/2) Z_S \otimes I + g N_\phi \otimes X_E$. Its two terms anticommute, so $H^2 = R(\epsilon)^2 I$ with $R(\epsilon) = \sqrt{g^2 + \epsilon^2/4}$. Its exact propagator is

$$e^{-ihH} = \cos(Rh) I - i \frac{\sin(Rh)}{R} H. \tag{6.12}$$

For any positive integer k, both residual frequencies

$$\epsilon_k^{\pm} = \pm\frac{2}{h}\sqrt{k^2\pi^2 - \chi^2} \tag{6.13}$$

give $Rh = k\pi$ and hence $e^{-ihH} = (-1)^k I$. The scalar sign cannot be observed. Every complete cell is the identity channel, independently of the nuisance direction.

Any controller restricted to cell boundaries consequently obtains the same complete data at two such resonant frequencies. This remains true with adaptation: equal input states and records lead through the same identity cell to equal boundary-instrument outcome laws; induction repeats the statement through every cell. Let the two frequencies be ω_1 and ω_2, separated by $D = |\omega_1 - \omega_2|$. Their common estimate distribution obeys

$$\frac{1}{2}\sum_{j=1}^{2}\mathbb{E}_{\omega_j}[(\widehat{\omega} - \omega_j)^2] = \mathbb{E}[(\widehat{\omega} - (\omega_1 + \omega_2)/2)^2] + \frac{D^2}{4} \geq \frac{D^2}{4}. \tag{6.14}$$

At least one frequency must have that much squared error, however many cells are used. This is an exact ambiguity, not merely a noisy estimate. Only resonances lying in the permitted frequency domain matter; its full Hamiltonian bound must also allow their strength $k\pi/h$.

The theorem below uses a fixed nonzero band near compensation where a simple calibration measurement is uniformly informative. The band does not shrink as the experiment grows. It is a sufficient identifiable band, not a claim of maximum bandwidth. The alias argument concerns boundary access at one fixed coupling and clock; it is not an obstruction proved for every possible inside-cell control.

7 How many resolved collisions improve an unknown frequency?

We now ask for an estimate rather than a yes-or-no decision. The frequency is initially unknown on a fixed interval. The aim is to make its largest mean-square estimation error small, even when the direction of the noise is also unknown. The discrimination theorem does not answer this question: a detectable difference between two devices need not provide a useful, unambiguous frequency reading. We need both an actual estimator and a bound that applies to every permitted estimator.

7.1 The fixed device and the two contact budgets

The sensor is a qubit S, with a protected ancillary qubit available for the constructions and arbitrary finite protected memory allowed in the comparison. Its frequency is ω. A fixed known compensation frequency ω_0 leaves the residual frequency $\epsilon = \omega - \omega_0$. Each interaction cell lasts $h > 0$. During a cell the sensor and its assigned bath qubit evolve under

$$H(\epsilon, \phi) = \frac{\epsilon}{2} Z_S \otimes I_E + g N_\phi \otimes X_E, \qquad N_\phi = \cos\phi\, X_S + \sin\phi\, Y_S. \tag{7.1}$$

The coupling $g > 0$ is fixed, and $\chi = gh$ lies in $(0, \pi/4)$. The same unknown direction ϕ applies throughout all cells and repeated trials. All bath qubits begin in $|0\rangle$ at time zero; after use they remain inaccessible and never interact again. Thus the construction uses new partners already present in the finite bath, not uncharged resets of an exposed environmental qubit.

There are n cells and a hard duration $T = nh$. At most $n-1$ contacts may act on the sensor strictly before the final time; at most m of them may occur strictly inside cells, where $0 \leq m \leq n-1$. Both restrictions hold on every classical history. Preparations at time zero and readout at T are

free. A readout earlier than T counts, even if its result is stored until T. Boundary operations performed at the same instant may be combined into one instrument. Neither feedback nor a frequency estimate changes the physical compensation, the coupling or the collision clock.

The counts have different jobs. In the resolution theorem, N counted all intermediate contacts in a fixed-time comparison over many devices. Here n counts the interaction cells of this particular sensor, and m counts contacts strictly inside them. Cell boundaries are not interior access, but operations there still consume the total contact allowance.

The comparison allows finite adaptive instrument trees, their parameter-independent classical mixtures, and limits whose event probabilities converge on an open parameter neighborhood while keeping the same structural budgets. Classical feedback may select real contact times. It does not grant coherent superpositions of times or continuous uncounted monitoring between contacts.

Choose a fixed interval $\Omega = [\omega_0 - E, \omega_0 + E]$, with $E > 0$ to be specified. Its width does not shrink as n grows. For an allowed procedure $\mathcal{P}$ and its numerical answer $\widehat{\omega}$, the risk is $\mathbb{E}^{\mathcal{P}}_{\omega,\phi}(\widehat{\omega} - \omega)^2$. The expectation averages every outcome, including unsuccessful records. The optimal worst-case risk is

$$R^*_{n,m}(\Omega) = \inf_{\mathcal{P},\widehat{\omega}} \sup_{\omega\in\Omega,\phi} \mathbb{E}^{\mathcal{P}}_{\omega,\phi}(\widehat{\omega} - \omega)^2. \tag{7.2}$$

The supremum asks for the most difficult allowed frequency and direction; the infimum selects the best complete experimental procedure. Biased estimates are allowed. Squared frequency error has units of inverse time squared.

Theorem 7.1 (Global precision with a contact allowance)**.** *For every fixed g, h with $0 < gh < \pi/4$, there is a fixed $E > 0$ such that*

$$\boxed{R^*_{n,m}(\Omega) = \Theta_{\chi,Eh}\left(\frac{1}{h^2(n+m^2)}\right)} \qquad (n \geq 1,\ 0 \leq m \leq n-1). \tag{7.3}$$

The multiplicative constants depend only on the fixed dimensionless hardware and band parameters χ, Eh, and are uniform in n, m, ϕ.

Here Θ means upper and lower bounds by positive constant multiples of the displayed expression. The theorem identifies an order of precision, not an optimal coefficient or a largest possible frequency band. When $m = 0$, the error has order $1/(h^2 n)$. An allowance of order $\sqrt{n}$ is the crossover scale. When m is proportional to n, the error has order $1/(h^2 n^2)$. We next build the upper bound, then explain why no permitted adaptive strategy improves its order. Ordinary repeated readings supply inverse-linear risk in n. Interior contacts let useful phase accumulate over a number of cells proportional to m, supplying inverse-square risk in m. We choose the better complete design. The expression $n + m^2$ does not assume that information from the two designs can simply be added.

7.2 One coupling supplies two temporal settings

Set $R = \sqrt{g^2 + \epsilon^2/4}$ and $r = Rh$. Here R is a scalar frequency scale; the protected quantum memory is a separate system. The two terms in (7.1) anticommute, so $H^2 = R^2 I$. Consequently its exact propagator is $\cos r\, I - i(\sin r/R)H$. Reading the two bath components gives the untouched, or plain, cell operators

$$K_{0,p} = \cos r\, I - i\frac{\epsilon}{2R}\sin r\, Z_S, \qquad K_{1,p} = -i\frac{g}{R}\sin r\, N_\phi. \tag{7.4}$$

The subscript 0 denotes no parity flip, and 1 denotes a flip. Although the bath is not measured, sensor–ancilla parity distinguishes these branches. For each branch operator K, its probability

operator $K^\dagger K$ is a scalar multiple of the identity. Thus each branch probability is independent of the incoming logical state.

For an echoed cell, insert one sensor Z_S pulse at its midpoint. The pulse reverses the coupling term and preserves the signal term. Thus it can undo noise while leaving a signal to measure. With both terms present, the off-compensation dynamics must still be calculated; the signal-free echo example alone does not settle it. A known final Z_S frame correction, performed at the endpoint, changes its propagator to

$$V = e^{-iH_- h/2} e^{-iH_+ h/2}, \qquad H_\pm = \frac{\epsilon}{2} Z_S \otimes I_E \pm g N_\phi \otimes X_E. \tag{7.5}$$

The frame correction is a real, known Z_S operation that cancels the extra Z_S left by the interior pulse. It is included in the endpoint instrument, rather than treated as a free change to the device. Multiplication of the two half-cell propagators yields

$$\begin{aligned} V &= (uI - ivZ_S) \otimes I_E - ixN'_\phi \otimes X_E, \\ u &= \cos r + \frac{g^2}{R^2}(1 - \cos r), \qquad v = \frac{\epsilon}{2R} \sin r, \qquad x = \frac{\epsilon g}{2R^2}(1 - \cos r), \end{aligned} \tag{7.6}$$

where $N'_\phi = \partial_\phi N_\phi$ and $u^2 + v^2 + x^2 = 1$. An odd outcome of this cell initially applies N'_ϕ rather than N_ϕ. Apply an additional known Z_S conditional on that outcome. Because $Z_S N'_\phi = -iN_\phi$, its corrected flip operator is $-xN_\phi$. The frame correction, parity measurement and conditional correction are parts of the same endpoint instrument. Only the midpoint pulse is an inside-cell contact.

Both settings now have the same unknown flip, up to a scalar, and a known rotation on the no-flip branch. Write

$$\begin{aligned} p_e &= x^2, & p_p &= \frac{g^2}{R^2} \sin^2 r, & q_j &= 1 - p_j, \\ \theta_e(\epsilon) &= 2 \arctan(v/u), & \theta_p(\epsilon) &= 2 \arctan\left(\frac{\epsilon}{2R} \tan r \right), \end{aligned} \tag{7.7}$$

so $K_{0,j} = \sqrt{q_j}\, e^{-i\theta_j Z_S/2}$ for $j = e, p$. We restrict the band so $r < \pi/4$; then u and $\cos r$ are positive and these are smooth, single-valued real phase functions. At compensation,

$$p_e(0) = 0, \qquad p_p(0) = \sin^2 \chi, \qquad A := \theta'_e(0) = \frac{\sin \chi}{g}, \qquad d_p := \theta'_p(0) = \frac{\tan \chi}{g}. \tag{7.8}$$

Primes on scalar functions mean differentiation with respect to residual frequency; A and d_p therefore have units of time. The prime on N'_ϕ instead denotes an angle derivative. The zero echo-flip probability at $\epsilon = 0$ does not make an echoed cell noiseless at every frequency: x is generally nonzero away from compensation.

7.3 The recorded path cancels the unknown direction

Encode a logical qubit using $|0_L\rangle = |00\rangle$ and $|1_L\rangle = |11\rangle$, and start a block in $|+_L\rangle = (|0_L\rangle + |1_L\rangle)/\sqrt{2}$. A parity measurement reports whether the two physical bits agree, without asking which of the two code states is present. Denote even parity by $\sigma = +1$ and odd parity by $\sigma = -1$. Use an echoed cell after an even result and a plain cell after an odd result. Keep every result. The odd-state correction above aligns the two flip operators; it does not return an odd state to even parity.

For a block of L cells, the record is $\mathcal{R} = (\sigma_0, \ldots, \sigma_L)$ with $\sigma_0 = +1$. If it ends even, its normalized logical output is exactly

$$|\psi_{\mathcal{R}}(\epsilon)\rangle = e^{-i\Phi_{\mathcal{R}}(\epsilon)Z_L/2}|+_L\rangle,$$
$$\Phi_{\mathcal{R}}(\epsilon) = \sum_{i=1}^{L} \sigma_{i-1}\mathbf{1}_{\{\sigma_i=\sigma_{i-1}\}}\theta_{j(\sigma_{i-1})}(\epsilon), \qquad j(+1) = e, \quad j(-1) = p. \tag{7.9}$$

Here Z_L acts as $+1$ on $|0_L\rangle$ and -1 on $|1_L\rangle$; the indicator is one only if that cell did not flip parity. The logical operators X_L, Y_L, Z_L are the Pauli matrices (1.2) in the ordered basis $|0_L\rangle, |1_L\rangle$. For example, X_L exchanges those two code states and $Y_L|0_L\rangle = i|1_L\rangle$. The same calculation as for the qubit clock (1.5) gives $\langle X_L\rangle = \cos\Phi_{\mathcal{R}}$ and $\langle Y_L\rangle = \sin\Phi_{\mathcal{R}}$. These are the two logical readout axes used below. The explanation is algebraic: $N_\phi^2 = I$, while $N_\phi Z_S N_\phi = -Z_S$. Moving a no-flip rotation past a preceding flip reverses its phase sign. An even number of the remaining flip factors then cancels completely. Scalar branch amplitudes determine the record's probability but do not alter its normalized logical state.

Thus a record gives a known *function* of the candidate frequency, not an observed value of the unknown frequency. For example, a record that stays even for three cells has $\Phi = 3\theta_e$. A record that flips into the odd sector, remains there for one plain cell, and then flips back contributes $-\theta_p$ from that intervening no-flip cell. The functions and their signs are calculable without knowing ϕ. Their probabilities still depend on ϵ; the estimator below keeps that dependence rather than replacing it by its value at zero.

For a complete four-cell example, follow this recorded path:

Cell	Starting parity	Setting	Result	Phase added
1	Even	Echo	Stay	$+\theta_e$
2	Even	Echo	Flip	0
3	Odd	Plain	Stay	$-\theta_p$
4	Odd	Plain	Flip	0

Write $U_j = e^{-i\theta_j Z_S/2}$ for a no-flip rotation. Omitting scalar branch amplitudes, the total operation is $N_\phi U_p N_\phi U_e = U_p^{-1} U_e$. The unknown direction has cancelled, and the phase is $\theta_e - \theta_p$. Its slope at zero is $A - d_p < 0$, so this even-ending record will fail the positive-slope test below. Four even stays instead give phase $4\theta_e$ and slope $4A$. The flipping path is possible away from compensation; at exact compensation its echo flip has zero probability.

7.4 Select records with useful sensitivity over a whole band

A useful record must have a phase that changes steadily with frequency, with slope proportional to the block length. The argument below shows that the mean accumulated slope grows like L, while its usual fluctuations grow only like $\sqrt{L}$. A threshold below the mean therefore retains a fixed positive fraction of long blocks. The frequency band is chosen so that a record passing this test at the nominal frequency remains sensitive at every candidate frequency.

An even record alone need not have a large net phase derivative: positive and negative contributions could cancel. We therefore use an observable record criterion. Choose $E > 0$ small enough that, throughout $|\epsilon| \le E$,

$$r \le r_* := \sqrt{\chi^2 + (Eh/2)^2} < \pi/4, \qquad 0 \le p_e < p_p < \tfrac{1}{2},$$
$$\mu^0(\epsilon) := \frac{p_p q_e A - p_e q_p d_p}{p_e + p_p} \ge \frac{3A}{4}, \qquad \max\{|\theta_e' - A|, |\theta_p' - d_p|\} \le \frac{A}{4}. \tag{7.10}$$

These conditions hold on a positive-width interval: at zero, $p_p > 0$, $p_e = 0$ and $\mu^0 = A$, while the phase derivatives are continuous. Fix such an interval once. Any separate bound on the complete interaction must cover $R \le r_*/h$ on it.

Replace each phase derivative in (7.9) by its value at zero, obtaining the known number

$$D_L^0 = \sum_{i=1}^{L} \sigma_{i-1} \mathbf{1}_{\{\sigma_i = \sigma_{i-1}\}} d_{j(\sigma_{i-1})}, \qquad d_e = A, \quad d_p = \frac{\tan \chi}{g}. \tag{7.11}$$

Accept a record precisely when $\sigma_L = +1$ and $D_L^0 \ge AL/2$. This rule uses measured signs and fixed calibration constants. It does not assume the true residual frequency is zero.

Here is why a fixed positive fraction of blocks pass. The parity chain has transition matrix

$$P = \begin{pmatrix} q_e & p_e \\ p_p & q_p \end{pmatrix}, \qquad \pi_+ = \frac{p_p}{p_e + p_p}, \qquad \lambda = 1 - p_e - p_p \le \rho < 1. \tag{7.12}$$

The final inequality holds uniformly by compactness and $p_p > 0$. Rows and columns of P are ordered even, odd. Its entry in a given row and column is the probability of moving between those two parities. The number π_+ is the long-run even fraction, and λ controls the decay of the initial parity imbalance. The exact recursion starting even gives $\Pr(\sigma_i = +1) = \pi_+ + (1 - \pi_+)\lambda^i \ge 1/2$. The stationary average derivative reward is μ^0; starting even adds a nonnegative transient contribution. Hence $\mathbb{E} D_L^0 \ge 3AL/4$.

For classical random variables, variance and covariance mean

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2], \qquad \operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]. \tag{7.13}$$

Variance measures the spread of an outcome around its mean; covariance measures whether two fluctuations tend to have the same sign. The variance tail inequality is $\Pr(|X - \mathbb{E}X| \ge t) \le \operatorname{Var}(X)/t^2$ for $t > 0$: on that event the squared fluctuation is at least t^2.

Let X_i be the ith summand of D_L^0. Rewards in nearby cells are correlated, but their correlations decay. Put $M = \max\{A, d_p\}$ and $V_0 = M^2[1 + 2/(1 - \rho)]$. For rewards X_i, X_j with $i < j$, conditioning on the parity after cell i leaves a difference between the two possible future mean rewards of at most $2M\rho^{j-i-1}$. A variable of range length $2M$ has variance at most M^2. Cauchy–Schwarz therefore gives $|\operatorname{Cov}(X_i, X_j)| \le M^2 \rho^{j-i-1}$, and summing the geometric series gives $\operatorname{Var}(D_L^0) \le V_0 L$. The elementary variance tail bound now implies

$$\Pr(\text{accepted}) \ge \frac{1}{2} - \frac{16 V_0}{A^2 L} \ge \frac{1}{4} =: p_0 \quad \text{if } L \ge L_0 := \max\{1, \lceil 64 V_0 / A^2 \rceil\}. \tag{7.14}$$

We subtracted the probability of failing the derivative test from the probability of ending even; these two events need not be independent. This proof also covers $p_e = 0$: an initially even chain is then absorbing, not an irreducible chain requiring a positive echo-flip probability.

For each accepted record, (7.10) bounds the change in its summed derivative by $AL/4$. Thus, at *every candidate frequency* $v \in [-E, E]$,

$$\alpha L \le \Phi'_{\mathcal{R}}(v) \le \beta L, \qquad \alpha = A/4, \qquad \beta = \max_{j=e,p} \sup_{|v| \le E} |\theta_j'(v)|. \tag{7.15}$$

The retained record certifies a whole increasing phase curve. Rejected records are flagged and charged the same block duration.

7.5 Paid calibration gives a first frequency interval

A plain one-cell trial already identifies frequency on this band. Prepare $|+_L\rangle$, measure parity, and on an even result measure Y_L. Record a classical score $z = +1$ or -1 for that logical readout, and $z = 0$ for odd parity. This score is not the Pauli operator Z_S. With $b = \epsilon \sin r/(2R)$, its exact likelihood is

$$\Pr(z = 0) = p_p, \qquad \Pr(z = \pm 1) = \tfrac{1}{2}(\cos r \pm b)^2, \qquad F(\epsilon) := \mathbb{E}z = \frac{\epsilon \sin(2Rh)}{2R}. \tag{7.16}$$

It is independent of ϕ. For instance, at $\chi = \pi/6$ and $\epsilon = 0$, the probabilities of $0, +1, -1$ are $1/4, 3/8, 3/8$. The zero scores are part of the data, not discarded shots. Differentiation gives

$$F'(\epsilon) = h\left[\frac{g^2}{R^2}\frac{\sin(2r)}{2r} + \frac{\epsilon^2}{4R^2}\cos(2r)\right] \in [\kappa, h], \qquad \kappa = h\cos(2r_*) > 0. \tag{7.17}$$

The two weights sum to one, and $\sin x/x \geq \cos x$ on $(0, \pi/2)$. Therefore F has a unique inverse on the promised band.

For N_c independent trials, average the scores, clip their mean to $F([-E, E])$, and apply F^{-1}. Call the result $\widehat{\epsilon}_c$. Clipping cannot increase its distance from the true mean, so

$$\begin{aligned} \mathbb{E}(\widehat{\epsilon}_c - \epsilon)^2 &\leq \frac{1}{N_c\kappa^2}, \\ \Pr(|\widehat{\epsilon}_c - \epsilon| \geq \delta) &\leq 2\exp[-N_c\kappa^2\delta^2/2]. \end{aligned} \tag{7.18}$$

The first claim uses $\mathrm{Var}(z) \leq 1$; the second is the bounded-variable concentration inequality for independent scores in $[-1, 1]$. Each trial costs one full cell. An intermediate readout and the next preparation share one boundary instrument. A calibration prefix of N_c cells costs N_c boundary contacts, because its last readout occurs before the full experiment's deadline.

7.6 Resolve phase ambiguity with nested intervals

A long-lived phase is not itself a frequency estimate: many frequencies can differ by whole turns of the phase. Use short blocks first, then double the block length as the allowed interval narrows. Each measurement is oriented relative to the phase predicted at the current interval center. Only the terminal measurement basis changes; (7.1) remains fixed.

The algorithm has a simple loop. First use short trials to obtain a safe interval. At each later stage, double the block length. For each observed record, calculate the predicted readout mean at every frequency still in the interval. Average those predicted curves, compare with the actual readout average, and invert the resulting curve. Keep a narrower interval around that answer, intersected with the previous interval. Earlier mistakes could cause larger errors, so early stages receive more repetitions. The constants below make this rule quantitative.

The following explicit schedule also controls rare mistakes. Set

$$s = \frac{\pi}{8\beta}, \quad L_k = 2^k L_0, \quad c_* = \min\left\{\frac{p_0^2}{2}, \frac{p_0\alpha^2 s^2}{8}\right\} > 0. \tag{7.19}$$

The quantity s has units of frequency; αs and βs are dimensionless. Fix a final stage $K \geq 1$. Use

$$N_c(K) = \left\lceil \frac{2L_0^2}{\kappa^2 s^2}(K\log 8 + \log 2)\right\rceil \tag{7.20}$$

calibration trials, and form $C_0 = [-E, E] \cap [\widehat{\epsilon}_c - s/L_0, \widehat{\epsilon}_c + s/L_0]$. Equation (7.18) bounds the chance that this misses the truth by 8^{-K}. It is always nonempty because the coarse answer is clipped. All logarithms in this schedule are natural.

At stage k, let c be the midpoint of C_{k-1} and attempt

$$r_k = \left\lceil \frac{(K-k)\log 8 + \log 12}{c_*} \right\rceil \tag{7.21}$$

independent blocks of length L_k. Each begins with a fresh $|+_L\rangle$ preparation and uses its own preinitialized bath partners. Within the stage, the parity-based cell policy stays unchanged, so earlier terminal readouts do not change the law of later block records. For each accepted record $\mathcal{R}_i$, measure $\cos[\Phi_{\mathcal{R}_i}(c)]Y_L - \sin[\Phi_{\mathcal{R}_i}(c)]X_L$. Its outcome $y_i \in \{-1, +1\}$ has conditional mean $\sin[\Phi_{\mathcal{R}_i}(\epsilon) - \Phi_{\mathcal{R}_i}(c)]$. If G blocks pass, their known mean curve is

$$H_k(v) = \frac{1}{G} \sum_{i \text{ accepted}} \sin[\Phi_{\mathcal{R}_i}(v) - \Phi_{\mathcal{R}_i}(c)], \qquad v \in C_{k-1}. \tag{7.22}$$

If $G < p_0 r_k/2$, use the fallback $\widetilde{\epsilon}_k = c$. Otherwise clip the observed average of the y_i to $H_k(C_{k-1})$ and invert H_k, obtaining $\widetilde{\epsilon}_k \in C_{k-1}$. In either case set

$$C_k = C_{k-1} \cap [\widetilde{\epsilon}_k - s/L_k, \widetilde{\epsilon}_k + s/L_k]. \tag{7.23}$$

Every interval remains nonempty and nested, even after an error. A one-point interval requires no inversion.

Why is the inverse unique? The preceding interval has radius at most s/L_{k-1} about c, so (7.15) gives

$$|\Phi_{\mathcal{R}_i}(v) - \Phi_{\mathcal{R}_i}(c)| \le \beta L_k \frac{s}{L_{k-1}} = \frac{\pi}{4}, \qquad H_k'(v) \ge \frac{\alpha L_k}{\sqrt{2}}. \tag{7.24}$$

The centered phase stays within one monotone sine branch. The full phase may have made many turns; earlier stages have already selected which turn is relevant.

The statistical argument must condition on the *actual records*. First, the fresh blocks have independent acceptance indicators, each with probability at least p_0. Thus the low-count probability is at most $e^{-p_0^2 r_k/2}$. Next hold all their records fixed. The accepted readouts are independent and their mean is exactly $H_k(\epsilon)$, although the probability of receiving those records depends on ϵ. The inverse slope above means an error larger than s/L_k requires a readout-average deviation larger than $\alpha s/\sqrt{2}$. On the event $G \ge p_0 r_k/2$, its probability is at most $2e^{-p_0 r_k \alpha^2 s^2/8}$. Averaging over records therefore gives

$$\Pr\{\epsilon \notin C_k \mid \text{earlier data with } \epsilon \in C_{k-1}\} \le 3e^{-c_* r_k} \le \tfrac{1}{4}\, 8^{-(K-k)}. \tag{7.25}$$

No assumption of frequency-independent record probabilities entered this calculation.

Return the midpoint of C_K. If no stage fails, the error is at most s/L_K. If the first failure is stage k, the truth and all later answers remain in the preceding interval, so the final error is at most $4s/L_k$. A coarse failure can cause error at most $2E$. These bounds yield

$$\begin{aligned} \mathbb{E}(\widehat{\epsilon} - \epsilon)^2 &\le \frac{s^2}{L_K^2} + 4E^2\, 8^{-K} + \sum_{k=1}^{K} \frac{16 s^2}{L_k^2} \frac{8^{-(K-k)}}{4} \\ &\le \frac{9s^2 + 4E^2 L_0^2}{L_K^2}. \end{aligned} \tag{7.26}$$

Longer blocks are repeated fewer times because their possible errors are smaller. This taper avoids an extra logarithmic cost. The deterministic number of cells is

$$B_K = N_c(K) + \sum_{k=1}^{K} r_k L_k \le C_* L_K, \tag{7.27}$$

with a fixed constant C_*. Indeed, $\sum_k L_k \le 2L_K$ and $\sum_k (K-k)L_k \le 2L_K$, while $N_c(K) = O(K+1) = O(2^K)$. Choose the largest schedule that fits an assigned cell budget Q. Since $Q < B_{K+1} \le 2C_* L_K$, (7.26) is $O(1/Q^2)$, uniformly on the band. Rescaling frequency to $h\epsilon$ writes this as $C_1/(h^2Q^2)$. A fixed clipped answer covers the finitely many smaller budgets after increasing C_1.

7.7 Charge the complete experiment

For $n \ge 3$ and $m \ge 1$, allocate

$$Q = \min\left\{m, \left\lfloor \frac{n-1}{2} \right\rfloor\right\} \tag{7.28}$$

cells to the entire estimator, including calibration and every rejected block. Each used cell has at most one midpoint pulse and one endpoint instrument. That endpoint combines frame correction, parity, conditional correction, any block readout and the next preparation. Count the last readout because it is early. Therefore

$$\#\text{ inside-cell contacts} \le Q \le m, \qquad \#\text{ all contacts} \le 2Q \le n-1. \tag{7.29}$$

Store the classical estimate through the remaining charged cells. Every history still costs the prescribed nh duration. Since $Q \ge m/3$, this gives risk at most $9C_1/(h^2m^2)$.

There is also the plain-cell estimator using all n cells. Its last readout occurs at T, so it uses exactly $n-1$ boundary contacts and no inside-cell contacts, with risk at most $C_0/(h^2n)$, where $C_0 = 1/\cos^2(2r_*)$. Choose whichever complete design has the smaller known bound. The inequality

$$\min\left\{\frac{C_0}{n}, \frac{9C_1}{m^2}\right\} \le \frac{C_0 + 9C_1}{n+m^2} \tag{7.30}$$

proves the required upper order. The cases $m = 0$ and $n = 1, 2$ use the plain-cell design. We have not spent n calibration cells and then added extra sensing cells for free.

7.8 Why arbitrary adaptive contacts cannot do better in order

For a lower bound, it is enough to exhibit a difficult one-parameter subfamily of the allowed unknowns. Let the residual frequency vary and choose a smooth nuisance curve $\phi(\epsilon)$ solely for this proof. The curve compares different possible worlds. Within each world, the noise direction remains fixed throughout the experiment; it is not drifting in physical time. A procedure that must work for every frequency and direction must also work on this difficult subset. The actual controller is not told that the parameters lie on this curve. For the plain cell write

$$K_0 = aI - ibZ_S, \qquad K_1 = -icN_\phi, \qquad a = \cos r, \quad b = \frac{\epsilon}{2R}\sin r, \quad c = \frac{g}{R}\sin r. \tag{7.31}$$

With $p = c^2$, $q = a^2 + b^2$ and $s_1 = ab' - ba' = q\theta_p'/2$, choose

$$\phi'(\epsilon) = \frac{s_1}{p} = \frac{q_p \theta_p'}{2p_p}. \tag{7.32}$$

This is smooth because $p_p \geq p_{\min} := \chi^2(\sin r_*/r_*)^2 > 0$. A dot now includes differentiation of both ϵ and $\phi(\epsilon)$. Using $N_\phi N'_\phi = iZ_S$ gives

$$\sum_{r=0}^{1} K_r^\dagger \dot{K}_r = i(-s_1 + p\phi')Z_S = 0, \qquad \sum_{r=0}^{1} \dot{K}_r^\dagger \dot{K}_r = \left(a'^2 + b'^2 + c'^2 + \frac{s_1^2}{p}\right) I. \tag{7.33}$$

The first identity removes coherent addition of derivatives across plain cells. At fixed direction, the time-integrated Hamiltonian derivative has norm at most $h/2$, hence $a'^2 + b'^2 + c'^2 \leq h^2/4$. Since $s_1^2 \leq q(a'^2 + b'^2)$, the derivative norm of a plain-cell isometry is at most

$$a_0 = \frac{h}{2\sqrt{p_{\min}}}. \tag{7.34}$$

An isometry is a map W satisfying $W^\dagger W = I$: it preserves lengths and inner products, possibly while embedding vectors in a larger space. Here it stores the two Kraus labels in a purification register. It is a proof device, not permission for the experimenter to measure the bath.

The zero derivative overlap has a concrete consequence. For two untouched cells with isometries W_1, W_2 and a fixed input $|\psi\rangle$, the two derivative contributions are

$$|d_1\rangle = W_2 \dot{W}_1 |\psi\rangle, \qquad |d_2\rangle = \dot{W}_2 W_1 |\psi\rangle. \tag{7.35}$$

Their inner product is

$$\langle d_1 | d_2 \rangle = \langle \psi | \dot{W}_1^\dagger W_2^\dagger \dot{W}_2 W_1 | \psi \rangle = 0, \tag{7.36}$$

because $W_2^\dagger \dot{W}_2 = 0$. Their squared lengths add, as for perpendicular sides of a right triangle. These are directions in the mathematical state space, not spatial noise directions. The full argument below preserves this property through adaptive operations. For contacted cells, the derivative vectors need not be perpendicular: their lengths can add before squaring. This distinction produces the scales $\sqrt{n}$ and m, and hence the two terms in the risk law.

For an active cell, meaning one containing a system contact strictly inside its interval, use the full joint evolution instead. Let $K_\phi = \sup_{|\epsilon| \leq E} |\phi'(\epsilon)|$. Its Hamiltonian derivative is

$$\dot{H} = \tfrac{1}{2} Z_S \otimes I_E + g\phi' N'_\phi \otimes X_E, \qquad \|\dot{H}\| = \sqrt{\tfrac{1}{4} + g^2 \phi'^2}. \tag{7.37}$$

The two terms anticommute. Integrating this bound over the cell gives $b_0 = h\sqrt{1/4 + g^2 K_\phi^2}$ for the derivative norm, even when several adaptive operations occur within it. Instrument rules and their outcome-dependent times are held fixed as functions of the recorded data when the unknown parameters are differentiated.

To see exactly where the hard contact budget enters, purify the complete experiment, retaining copies of classical activity flags for analysis. The memory-only normal form places each flag at cell entrance, without moving any system contact. Differentiation of the circuit gives a sum $\partial_\epsilon |\Psi\rangle = \sum_j (C_j + D_j)$: C_j is the derivative inserted in inactive cell j, and D_j the derivative inserted in an active cell. Common future isometries preserve inner products. The zero identity in the plain-cell calculation makes distinct C_j orthogonal, so

$$\left\| \sum_j C_j \right\|^2 \leq n a_0^2. \tag{7.38}$$

For the active terms, let Q_z project onto the archived complete activity history $z = (z_1, \ldots, z_n)$, and let π_j be the probability that cell j is active. Since $Q_z D_j = 0$ when $z_j = 0$ and every history

has at most m active cells, Cauchy–Schwarz within each history gives

$$\left\|\sum_j D_j\right\|^2 = \sum_z \left\|\sum_{j:z_j=1} Q_z D_j\right\|^2 \\ \leq m\sum_j \|D_j\|^2 \leq mb_0^2\sum_j \pi_j \leq m^2 b_0^2. \tag{7.39}$$

One must count histories this way; summing separate square roots of activity probabilities loses the m dependence. The budget is a structural rule even on histories of zero probability at a particular frequency. Copying its already classical flags for the proof does not supply extra observations to the controller.

For a differentiable classical outcome distribution $p_\epsilon(x)$, its Fisher information is $J_\epsilon = \mathbb{E}[(\partial_\epsilon \log p_\epsilon(X))^2]$. It measures how quickly nearby probability distributions separate locally; it is not itself a global error. Every accessible measurement has information at most four times the squared derivative norm of a purification. The triangle inequality and (7.39) imply, uniformly throughout the band,

$$J_\epsilon \leq 4(a_0\sqrt{n} + b_0 m)^2 \leq 4(a_0^2 + b_0^2)(n + m^2). \tag{7.40}$$

This converse may grant arbitrary boundary operations as a relaxation; it still enforces the actual inside-cell budget on every branch. Discarding the inaccessible purification cannot increase the information.

7.9 Turn the local bound into a global risk obstruction

A reciprocal Fisher-information formula alone would require conditions that do not establish our global claim. Instead average over the fixed probability density

$$\pi_E(u) = \frac{1}{E}\cos^2\left(\frac{\pi u}{2E}\right), \qquad -E < u < E, \qquad I(\pi_E) = \int \frac{\pi_E'(u)^2}{\pi_E(u)}\,\mathrm{d}u = \frac{\pi^2}{E^2}. \tag{7.41}$$

This density is a lower-bound device, not prior knowledge assumed for the experiment. Put the physical parameters on (7.32). For the joint density of u and the complete outcome x, let $S(u,x) = \partial_u \log[\pi_E(u)p_u(x)]$. Integration by parts, with the prior vanishing at the endpoints, gives

$$\mathbb{E}[(\widehat{\epsilon}(X) - u)S(u,X)] = 1, \qquad \mathbb{E}S^2 = \mathbb{E}_{\pi_E} J_u + I(\pi_E). \tag{7.42}$$

For a fixed outcome x, the reported estimate is a fixed number, so $\partial_u[\widehat{\epsilon}(x) - u] = -1$. The boundary term vanishes because the prior is zero there; integrating that derivative leaves the normalized joint density and gives one. Cauchy–Schwarz now yields

$$1 \leq \mathbb{E}[(\widehat{\epsilon} - u)^2]\,\mathbb{E}[S^2]. \tag{7.43}$$

It therefore bounds the average squared error from below by the reciprocal of the second expression. Worst-case error is at least this average error. No unbiasedness condition was imposed. Substitution of (7.40) proves

$$R^*_{n,m}(\Omega) \geq \frac{1}{4(a_0^2 + b_0^2)(n + m^2) + \pi^2/E^2}. \tag{7.44}$$

Because $a_0/h, b_0/h, Eh$ are fixed, this is the lower order in (7.3).

The stipulated limits of finite adaptive controllers retain this lower bound without assuming continuity of Fisher information. Clip any final estimate to the frequency band and round it

to a finite mesh of spacing at most η. Its outcome events form a fixed finite partition. Event-probability convergence transfers the averaged risk of this rounded estimate; the loss is bounded by $4E^2$. Rounding changes squared error by at most $4E\eta$, so sending η to zero transfers the same lower bound to the original estimate. This is why the full global conclusion survives the operational closure.

Temporal access has supplied two distinct conditional evolutions at one fixed coupling. Their retained records give uniformly usable phases, and the fully paid multiscale estimator turns those phases into a global answer. The converse limits every allowed adaptive contact strategy, not merely the chosen echo construction. The attained bound still assumes ideal midpoint pulses and the specified fixed band. Timing uncertainty changes which conditional states are actually available, which is the next question.

8 The information carried by pulse timing

The ideal-contact result used a pulse exactly halfway through a collision. How much timing accuracy does that conclusion require? Before assigning a number to the error, we must specify what the controller knows. Imagine that one collision lasts 100 clock ticks, with its midpoint at tick 50.

Timing model	Information available to the controller
Known fixed displacement	Every pulse arrives at tick 55, and this fact is supplied.
Unknown fixed displacement	Every pulse has the same offset, whose value must first be learned.
Hidden independent jitter	Each pulse independently arrives at tick 45 or 55; their probabilities are known, but the realized choice is never reported.

The ticks illustrate the distinction; they do not provide an extra timing measurement. A known displacement changes a definite quantum operation. A hidden random displacement averages several operations. These are different experiments even when their maximum errors have the same magnitude.

Keep the sensor, coupling g, cell duration h, fixed compensation ω_0, and shared unknown direction ϕ from the preceding sensing task. Thus $\epsilon = \omega - \omega_0$ is the residual frequency and $\chi = gh \in (0, \pi/4)$. In this section the allowed menu consists of plain cells and cells containing exactly one sensor Z_S pulse. In a pulsed cell there is no further interior contact, no choice among extra pulse positions, and no change of physical compensation. The choice between plain and pulsed cells may depend on the earlier classical record. Boundary instruments and protected memory remain available. The hard budgets are still $n-1$ total nonterminal contacts and at most m interior pulses over duration nh.

8.1 Derive the displaced cell before averaging it

Place the pulse at $h/2 + \delta$, where $|\delta| < h/2$. The simultaneous Hamiltonians on the two sides of the pulse, after accounting for its known frame, are

$$H_{\pm} = \frac{\epsilon}{2} Z_S \otimes I_E \pm g N_\phi \otimes X_E, \qquad R = \sqrt{g^2 + \epsilon^2/4}. \tag{8.1}$$

Here $N_\phi = \cos\phi\, X_S + \sin\phi\, Y_S$ and $N'_\phi = \partial_\phi N_\phi = -\sin\phi\, X_S + \cos\phi\, Y_S$. A prime on this operator means an angle derivative; primes on scalar coefficients below mean frequency derivatives. Both N_ϕ and N'_ϕ are Hermitian unitaries, and they anticommute.

The corrected joint propagator is exactly

$$V_\delta = e^{-iH_-(h/2-\delta)} e^{-iH_+(h/2+\delta)}. \tag{8.2}$$

The physical pulse frame is corrected within the already counted boundary instrument or the final readout. No additional free sensor pulse is hidden in this notation. Each bath qubit is initialized at time zero, interacts in its one assigned cell, and is retained without later access or reuse.

Because $H_\pm^2 = R^2 I$, expand each exponential as $\cos(Rt)I - i\sin(Rt)H_\pm/R$ and multiply. Define the real, dimensionless coefficients

$$\begin{aligned} D &= \cos(2R\delta) - \cos(Rh), & u &= \cos(Rh) + \frac{g^2}{R^2} D, \\ v &= \frac{\epsilon}{2R}\sin(Rh), & w &= \frac{g}{R}\sin(2R\delta), \\ x &= \frac{\epsilon g}{2R^2} D. \end{aligned} \tag{8.3}$$

With bath input $|0\rangle$, the two parity-resolved Kraus operators are

$$K_0 = uI - ivZ_S, \qquad K_1 = -i(wN_\phi + xN'_\phi), \qquad u^2 + v^2 + w^2 + x^2 = 1. \tag{8.4}$$

For example, $(wN_\phi + xN'_\phi)^2 = (w^2 + x^2)I$ because the cross terms cancel. Consequently each observed parity branch is proportional to a unitary. Parity has learned which branch occurred without producing an additional unobserved mixture inside that branch.

8.2 A supplied fixed displacement can preserve global precision

Suppose δ is fixed, known and nonzero, with $0 < |\delta| < h/2$. The assertion is that a plain/displaced-pulse menu still attains inverse-square global risk on a sufficiently small fixed frequency band when m is proportional to n. The band and constants may depend on this supplied displacement. Their uniformity as δ approaches zero or a cell endpoint is not asserted. The exactly centered case is already covered by the ideal-contact construction.

Near $\epsilon = 0$, w has a fixed nonzero sign. Absorb that sign into an irrelevant scalar phase and write

$$\theta_e(\epsilon) = 2\arctan(v/u), \qquad \zeta_e(\epsilon) = \arctan(x/w). \tag{8.5}$$

Subscript e denotes the displaced pulsed setting, and p denotes the plain setting. Its no-flip branch rotates about Z_S by θ_e; its normalized flip is $N_{\phi+\zeta_e}$, up to a scalar phase. For a plain cell the corresponding extra angle is $\zeta_p = 0$. The angle ζ_e depends on the unknown frequency, but its *function* of frequency is known from the specified hardware. This is just the kind of known nonlinear dependence used by the recorded-phase estimator.

Use pulsed cells while parity is even and plain cells while it is odd. Let $\sigma_{i-1} \in \{+1, -1\}$ be parity before cell i, and let j_i be its selected setting. Commuting known Z rotations through successive flips shows that every record $\mathcal{R}$ ending even has the exact logical phase

$$\Phi_{\mathcal{R}}(\epsilon) = \sum_i \sigma_{i-1} \left[\mathbf{1}_{\text{stay}}\theta_{j_i}(\epsilon) - 2\mathbf{1}_{\text{flip}}\zeta_{j_i}(\epsilon)\right]. \tag{8.6}$$

The common unknown ϕ cancels in pairs. The known extra flip angles do not cancel in general; they must be included. Dropping their derivatives would give the wrong sensitivity calculation.

Here is a direct check that these phases have useful drift. Put $z = 2g\delta$, so $0 < |z| < \chi$. At compensation,

$$\begin{aligned} p_e &= \sin^2 z, & p_p &= \sin^2 \chi, \\ d_e := \theta_e'(0) &= \frac{\sin\chi}{g\cos z}, & z_e := \zeta_e'(0) &= \frac{\cos z - \cos\chi}{2g\sin z}, \\ d_p := \theta_p'(0) &= \frac{\tan\chi}{g}. \end{aligned} \tag{8.7}$$

These derivatives have units of time. The mean derivative reward conditional on starting in the even sector is $f_e = (1-p_e)d_e - 2p_e z_e$; the magnitude of the odd-sector reward is $f_p = (1-p_p)d_p$. Simple trigonometric rearrangement gives

$$\frac{f_e}{p_e} - \frac{f_p}{p_p} = \frac{(\sin\chi - \sin z)\sin(\chi+z)}{g\sin\chi\,\sin^2 z} > 0. \tag{8.8}$$

For $-\chi < z < \chi$, the two factors in the numerator are positive. The stationary even and odd probabilities are $p_p/(p_e+p_p)$ and $p_e/(p_e+p_p)$. Hence the mean phase derivative gained per cell is

$$\mu_\delta = \frac{p_p f_e - p_e f_p}{p_e + p_p} > 0. \tag{8.9}$$

The stronger return from the odd sector again gives a net forward signal.

To make a uniform estimator, compute from each length-L record the *frozen* reward

$$D_L^0 = \sum_{i=1}^{L} \sigma_{i-1}\left[\mathbf{1}_{\text{stay}}\theta_{j_i}'(0) - 2\mathbf{1}_{\text{flip}}\zeta_{j_i}'(0)\right]. \tag{8.10}$$

"Frozen" means that the coefficients are evaluated at the fixed nominal point, not at the unknown true frequency. The gate that accepts an even record with $D_L^0 \geq \mu_\delta L/2$ is therefore implementable.

Continuity allows one fixed band on which $p_e < p_p < 1/2$, the stationary mean of this frozen reward is at least $3\mu_\delta/4$, and the difference between its even- and odd-start conditional means is positive. Starting even then gives mean at least $3\mu_\delta L/4$. If M bounds one reward in magnitude and $\rho < 1$ bounds the nontrivial eigenvalue of the two-state parity chain, conditioning across the intervening transitions yields

$$\mathrm{Var}(D_L^0) \leq VL, \qquad V = M^2\left(1 + \frac{2}{1-\rho}\right). \tag{8.11}$$

The variance grows linearly because correlations decay geometrically. Chebyshev's inequality gives probability at most $16V/(\mu_\delta^2 L)$ of falling below the threshold. The final-even probability is at least $1/2$. Thus

$$\Pr\{\text{final even},\ D_L^0 \geq \mu_\delta L/2\} \geq \frac{1}{4}, \qquad L \geq \frac{64V}{\mu_\delta^2}. \tag{8.12}$$

Shrink the same fixed band so that every actual derivative reward differs from its frozen value by at most $\mu_\delta/4$. Every accepted record then satisfies, throughout that band,

$$\frac{\mu_\delta L}{4} \leq \Phi_{\mathcal{R}}'(\epsilon) \leq \beta L \tag{8.13}$$

for a fixed finite time constant β. This supplies precisely the recorded-phase conditions: an observable gate, a nonvanishing acceptance probability and uniformly useful phase sensitivity.

The conditional inversion, paid coarse calibration and multiscale schedule from the preceding section therefore apply. Allocate

$$B_{\text{act}} = \min\left\{m, \left\lfloor \frac{n-1}{2} \right\rfloor\right\} \tag{8.14}$$

active cells. They cost at most B_{act} interior pulses and B_{act} boundary instruments, including the last early readout. Rejected blocks, repetitions and localization all fit inside that allocation. Store the estimate until the charged horizon nh. If $m = \Theta(n)$, the risk has order n^{-2}; the signal-generator bound gives the matching lower order. No retiming is needed for this construction.

If a calibrated command offset is an additional permitted setting, shifting the command by $-\delta$ instead restores the midpoint. Neither argument learns an unknown displacement. That would require an identifiable calibration experiment, an accuracy analysis and a charged calibration cost. An estimate of an offset is not an exact physical timing correction.

8.3 Hidden jitter produces an unresolved mixture

Now let every pulsed cell independently use $\delta = +d$ or $-d$ with equal probability. The amplitude $0 < d < h/2$ is known; the signs are never reported. They are independent of the unknown parameters, previous records and chosen controls. The controller may choose plain or jittered at cell entrance, but cannot learn the timing sign or add another interior pulse.

Only w in (8.3) changes sign. Averaging the physical channels at $\delta = \pm d$ cancels the cross terms between w and x. An exact Kraus description is therefore

$$L_0 = uI - ivZ_S, \qquad L_1 = -iwN_\phi, \qquad L_2 = -ixN'_\phi, \tag{8.15}$$

with coefficients evaluated at $\delta = d$. Parity distinguishes the even branch L_0 from the combined odd branch, but it cannot distinguish L_1 from L_2. Conditional on an odd result, an input density matrix ρ_S becomes

$$\rho_{S,\mathrm{odd}} = \frac{w^2 N_\phi \rho_S N_\phi + x^2 N'_\phi \rho_S N'_\phi}{w^2 + x^2}. \tag{8.16}$$

The same formula applies with an identity on the protected reference. In general this is a mixture of two states, rather than a single pure phase state. Assigning a separate observed label to L_1 and L_2 would change the experiment.

Failure of the preceding estimator does not yet prove a limitation on all other strategies. The converse below supplies that missing step. Choose fixed $\chi < r_* < \pi/4$ and define

$$E = \frac{2}{h}\sqrt{r_*^2 - \chi^2}, \qquad \Omega = [\omega_0 - E, \omega_0 + E]. \tag{8.17}$$

This is a nonshrinking identifiable band; the full interaction norm on it is at most r_*/h. For each fixed nonzero $d < h/2$, the restricted menu obeys

$$R^*_{n,m}(d;\Omega) = \Theta(n^{-1}), \qquad 0 \le m \le n-1. \tag{8.18}$$

Constants may depend on g, h, E, d, but not on n, m or the shared direction. The upper order is already attainable by ignoring pulsed cells and using the plain-cell calibration design. Its risk is at most $[nh^2\cos^2(2r_*)]^{-1}$, with every outcome retained and all nh charged. It uses $n-1$ boundary contacts and no interior contacts.

8.4 A common nuisance curve bounds every adaptive strategy

To prove the lower order, it is sufficient to exhibit a difficult subset of the allowed parameters. Choose a fixed compact residual-frequency interval strictly inside $(0, E)$. For a plain cell write

$$a_p = \cos(Rh), \quad b_p = \frac{\epsilon}{2R}\sin(Rh), \quad c_p = \frac{g}{R}\sin(Rh), \quad p_p = c_p^2. \tag{8.19}$$

Define a mathematical nuisance curve by

$$\phi'(\epsilon) = k(\epsilon) = \frac{a_p b'_p - b_p a'_p}{p_p}. \tag{8.20}$$

It is smooth because $p_p > 0$. A dot below denotes differentiation along this curve, changing frequency and nuisance angle together. A prime on a scalar still means its ordinary derivative with respect to ϵ. The controller is not told that nature follows this curve.

For the two plain-cell Kraus operators, direct differentiation gives

$$\begin{aligned} &\sum_j K_j^\dagger \dot{K}_j = 0, \\ &\sum_j \dot{K}_j^\dagger \dot{K}_j = \big[(a'_p)^2 + (b'_p)^2 + (c'_p)^2 + p_p k^2\big] I. \end{aligned} \tag{8.21}$$

As in the contact converse, the zero overlap prevents coherent accumulation of successive derivative contributions in a purification. We need the same property for the selectable jittered cell *on the same curve*.

Put $p = w^2 + x^2$ and $a = uv' - vu'$. The initial three-Kraus description has

$$\beta := i \sum_j L_j^\dagger \dot{L}_j = (a - pk) Z_S, \qquad L_1^\dagger L_2 = iwx Z_S. \tag{8.22}$$

On the chosen positive interval, $w, x > 0$: indeed $0 < 2Rd < Rh < \pi/4$ and $\cos(2Rd) > \cos(Rh)$. The unresolved odd labels provide a useful freedom in the proof. A unitary change of the inaccessible environmental basis changes the Kraus operators but leaves their summed channel, and every accessible outcome probability, unchanged. This is called an *environment gauge*. It is bookkeeping in a purification, not a pulse or bath measurement available to the controller.

Take a Hermitian 3×3 matrix G acting on the Kraus labels, whose only nonzero entries are

$$G_{12} = i\eta_G, \qquad G_{21} = -i\eta_G, \qquad \eta_G = \frac{a - pk}{2wx} \tag{8.23}$$

Its entries have units of time. In the derivative convention $\widetilde{\dot{L}} = \dot{L} - iGL$, the three derivatives are

$$\begin{aligned} \widetilde{\dot{L}}_0 &= u' I - iv' Z_S, \\ \widetilde{\dot{L}}_1 &= -i\big[w' N_\phi + (wk + \eta_G x) N'_\phi\big], \\ \widetilde{\dot{L}}_2 &= -i\big[x' N'_\phi - (xk + \eta_G w) N_\phi\big]. \end{aligned} \tag{8.24}$$

Consequently

$$\begin{aligned} \widetilde{\beta} &= (a - pk - 2wx\eta_G) Z_S = 0, \\ \widetilde{\alpha} := \sum_j \widetilde{\dot{L}}_j^\dagger \widetilde{\dot{L}}_j &= \big[(u')^2 + (v')^2 + (w')^2 + (x')^2 \\ &\qquad + (wk + \eta_G x)^2 + (xk + \eta_G w)^2\big] I. \end{aligned} \tag{8.25}$$

This is a genuine smooth family of purifications: solve $U'_E = -iU_E G$ on the chosen interval and replace the Kraus column by $U_E L$. All coefficients are bounded on that compact interval. Let A_*, with units of time squared, bound both the scalar in (8.25) and the plain-cell scalar in (8.21).

Both selectable channels now have zero derivative overlap. Dilate all adaptive boundary instruments, retain their classical histories, and condition the setting choice on the corresponding

history. Derivative sources from different cells are orthogonal, so their squared norms add. The complete accessible output therefore has information at most

$$F_{\mathrm{out}} \le 4nA_*. \tag{8.26}$$

Arbitrary protected memory and adaptive setting choices are included. Granting unlimited boundary contacts only strengthens this converse; the actual attaining design still obeys the original contact budget.

For clarity, choose a prior centered at $\epsilon_c > 0$, of half-width $\ell > 0$, whose support stays in the positive interval:

$$\pi(\epsilon) = \frac{1}{\ell}\cos^2\!\left(\frac{\pi(\epsilon-\epsilon_c)}{2\ell}\right), \qquad |\epsilon-\epsilon_c| \le \ell. \tag{8.27}$$

It vanishes at its endpoints and its Fisher information is $\int(\pi')^2/\pi\,\mathrm{d}\epsilon = \pi^2/\ell^2$. The integration-by-parts risk inequality from the preceding section yields

$$R^*_{n,m}(d;\Omega) \ge \frac{1}{4nA_* + \pi^2/\ell^2}. \tag{8.28}$$

It permits biased estimators and includes every branch. Worst-case risk on Ω is at least this prior-averaged risk. The gauge becomes singular at compensation because $x = 0$ there, but the prior never reaches that point. Thus (8.28) is a fixed-band minimax obstruction, not a claim of inverse-linear risk at each individual frequency. Clipping and finite quantization pass the bounded-loss argument to the specified operational limits; continuity of Fisher information is not presumed. Together with plain-cell calibration this proves (8.18).

8.5 Why quadratic local information can still survive

Here *local* means the slope of outcome probabilities at one specified frequency, before requiring a useful estimate on a whole interval. With two unknown parameters, the classical Fisher matrix has entries $J_{ab} = \mathbb{E}[(\partial_a \log p)(\partial_b \log p)]$, where a, b range over ϵ, ϕ and p is the complete outcome law. If $J_{\phi\phi} > 0$, the frequency information remaining after allowing the nuisance angle to vary is $J_{\epsilon\epsilon} - J_{\epsilon\phi}^2/J_{\phi\phi}$. This is called efficient local information. If the angle has zero first-order score, there is no such subtraction.

Exactly at compensation, the third Kraus branch vanishes. The number of nonzero branches changes there, so an argument valid away from that point need not describe its local sensitivity. Define

$$c = \cos(2gd), \quad s = \sin(2gd), \quad p_e = s^2, \quad A = \frac{\sin\chi}{g}, \quad B = \frac{c-\cos\chi}{g}. \tag{8.29}$$

Then $L_0 = cI$, $L_1 = -isN_\phi$ and $L_2 = 0$. Their frequency derivatives at fixed ϕ are $-iAZ_S/2$, 0 and $-iBN'_\phi/2$. Although the last derivative is nonzero, its contribution to the density matrix starts quadratically:

$$L_2(\epsilon)\rho L_2(\epsilon)^\dagger = \frac{\epsilon^2 B^2}{4} N'_\phi \rho N'_\phi + O(\epsilon^3). \tag{8.30}$$

It is absent from the first derivative at zero. The first-order accessible instrument consequently has no-flip phase derivative $d_e = A/c$ and a frequency-independent flip at that point.

Use jittered cells when even and plain cells when odd, keeping parity records. With $p_p = \sin^2\chi$ and $d_p = \tan\chi/g$, the signed phase derivative gained per unit physical time is

$$\begin{aligned} v_d &= \frac{p_p(1-p_e)d_e - p_e(1-p_p)d_p}{h(p_p+p_e)} \\ &= \frac{A\left[\sin^2\chi\cos(2gd) - \sin^2(2gd)\cos\chi\right]}{h(p_p+p_e)} > 0. \end{aligned} \tag{8.31}$$

The function $\sin^2 z/\cos z$ increases strictly on $(0, \pi/4)$, and $0 < 2gd < \chi$, which proves positivity.

A local information statement still needs an actual readout and a paid finishing rule. For an allocation of N cells, put $q_p = 1 - p_p < 1$ and reserve

$$k_N = \left\lceil \frac{3 \log N}{-\log q_p} \right\rceil \tag{8.32}$$

cells, for sufficiently large N that $k_N < N$. Here log is the natural logarithm. Run the parity policy for the first $N - k_N$ cells. If parity is even, measure the logical Y_L quadrature immediately and store the outcome. If parity is odd, use plain cells until the first return to even, or until the allocation ends. Read Y_L on return; retain a failure flag if no return occurs. The failure probability at compensation is at most $q_p^{k_N} \le N^{-3}$. All reserved cells are part of the allocation, even when readout occurs early.

The accumulated phase derivative has mean $v_d N h + O(\log N)$ and fluctuations of order $\sqrt{N}$; the geometric mixing of the two-state chain gives its variance of order N. At an even endpoint the zero-frequency logical state is independent of ϕ. Conditional Y_L probabilities are $1/2$ at zero, with opposite derivatives equal to one half of the recorded phase slope. Their conditional classical Fisher information is therefore the squared slope. Averaging complete records and retaining the small failure branch gives

$$\mathcal{I}_{\mathrm{local}} = v_d^2 (Nh)^2 + o(N^2). \tag{8.33}$$

Here $o(N^2)$ denotes a correction whose ratio to N^2 tends to zero. The nuisance has no first-order score on these even-readout branches, so the same leading information survives elimination of the shared angle.

Take $N = \min\{m, \lfloor (n-1)/2 \rfloor\}$. There are at most N interior pulses and N boundary contacts, including the early readout. Store the result through the remaining charged cells. For $m = \Theta(n)$ this is quadratic local information under the original total-contact budget. It coexists with inverse-linear optimal global risk because a readout's slope at one special frequency need not control error uniformly on a fixed interval. This is a statement about pointwise classical scores of an explicit readout; it makes no continuity assumption about quantum Fisher information when a Kraus branch disappears.

8.6 What accuracy is necessary, and what remains unproved

The away-from-zero gauge also quantifies a necessary accuracy scale. Fix $d_{\max} < h/2$, keep the same positive prior interval, and put $\xi = d/h$. Uniformly for $0 < d \le d_{\max}$, w is at least a positive constant times ξ, while x stays bounded below by a positive constant. The other coefficients and their frequency derivatives stay bounded. Thus (8.23) grows at most as h/ξ, and the derivative costs satisfy

$$\alpha_{\mathrm{plain}} \le C_0 h^2 I, \qquad \widetilde{\alpha}_{\mathrm{jitter}} \le \frac{C_1 h^2}{\xi^2} I. \tag{8.34}$$

The constants depend on fixed dimensionless hardware and the prior, not on n, m, d. Orthogonality now charges the expected number of jittered cells, which is at most the hard branchwise bound m. Combining this estimate with the independent signal-generator limitation yields, for a fixed C,

$$R^*_{n,m}(d;\Omega) \ge \max\left\{ \frac{1}{n^2 h^2 + \pi^2/\ell^2}, \quad \frac{1}{C h^2 (n + m/\xi^2) + \pi^2/\ell^2} \right\}. \tag{8.35}$$

For $m = \Theta(n)$, the second denominator must grow at least quadratically if the risk is to be $O(n^{-2})$. This requires $1/\xi^2$ to grow at least linearly in n. Since $\xi = d/h$, attaining that risk would require

$$d = O(h/\sqrt{n}). \tag{8.36}$$

This is a necessary asymptotic order, not an exact finite-budget tolerance and not a sufficient condition. The proof supplies a limitation; it does not construct a globally optimal estimator for shrinking jitter or determine the best joint dependence on n, m, d.

A separate local-information comparison helps locate the role of the menu. In this auxiliary comparison, N counts uses of one repeated primitive, and its contact cost is stated separately; we are not imposing the earlier $N-1$ total-contact allowance over an N-cell horizon. If *every* exposed cell repeats the same jittered echo, with arbitrary boundary controls but no plain-cell option, its optimal efficient local information at compensation satisfies

$$\frac{N^2A^2}{1+(N-1)p_e} \le \mathcal{I}^*_{\mathrm{echo}}(N,d) \le \min\left\{N^2h^2, N\left(\frac{A^2}{p_e}+B^2\right)\right\} \qquad (d>0). \tag{8.37}$$

For the upper bound, the nuisance slope $k = cA/(2p_e)$ already cancels the three-Kraus derivative overlap at zero. For the lower bound, measure parity and apply the same sensor X_S recovery after each cell, then use a recorded uniformly randomized equatorial readout. At compensation the flip count K is binomial with parameters N, p_e. Its conditional phase-derivative vector is $((N-K)A/c, -2K)$: the first component differentiates frequency, and the second differentiates the shared angle. Pooling the full two-parameter Fisher matrices before eliminating ϕ gives the displayed lower expression. This implementation uses N interior pulses and $N-1$ boundary instruments, so it costs $2N-1$ nonterminal contacts over its own N-cell horizon. To fit this implementation into the original n-cell task, one may instead take $N \le \min\{m, \lfloor (n-1)/2 \rfloor\}$, count its last readout as an early contact, and store the answer until nh. The paid total is then $2N \le n-1$, with the unused duration still charged.

At $d=0$ the lower expression continues to N^2A^2; the signal cap remains N^2h^2. Since $A^2+B^2 \le h^2$ and $p_e = \Theta_\chi((d/h)^2)$, the single repeated primitive has

$$\mathcal{I}^*_{\mathrm{echo}}(N,d) = \Theta_\chi\left(\frac{h^2N^2}{1+N(d/h)^2}\right). \tag{8.38}$$

Its need for improving timing to preserve *local* quadratic order does not apply to the richer plain/jittered menu, whose local protocol was just constructed. A claim about timing must name the permitted controls as well as the error distribution and the available timing record.

The established boundary is therefore precise. Supplied systematic timing information can preserve the record's pure phase and global quadratic order. Hidden independent jitter mixes odd operations and imposes inverse-linear global risk for this one-pulse menu at fixed nonzero amplitude. Additional interior pulses, observed timing labels, selectable couplings, finite-duration controls and retuned compensation are different resources. The present converse does not evaluate them.

9 What temporal access changes

The first question was whether a memoryless law that reproduces a final channel must also reproduce an experiment that touches the device earlier. The answer is quantitative. For the centered, drift-free dissipators in the stated model, one intrinsic lifetime and a strength bound Λ give

$$\Gamma_N, \Delta_N = \Theta\left(\min\left\{1, N\frac{\nu^2}{\Lambda^2}\right\}\right). \tag{9.1}$$

The survival test supplies the universal direction: short-time Markov loss is linear while bounded microscopic leakage is quadratic. The exact collision simulator supplies the converse: untouched

cells match perfectly, interrupted cells have quadratic action error, and the physical-prefix argument charges only the cells touched on each realized history. One test works against every bath, and one simulator hides against every tester. Those two statements establish the contact scale together.

With no intermediate system contact before the final readout, the closest microscopic model is indistinguishable at the specified final time. At sufficiently many contacts, a fixed positive advantage becomes unavoidable. Neither statement says that the bath has been identified. The test does not reconstruct its Hamiltonian, determine its dimension, or establish whether its information storage is classical or quantum. It resolves an observable difference between the permitted descriptions.

The sensing application asks a different question about a particular microscopic device. With n fixed-duration cells and at most m contacts inside them, within $n-1$ total nonterminal contacts, the optimal worst-case frequency error is

$$R^{*}_{n,m} = \Theta_{\chi,Eh}\left(\frac{1}{h^{2}(n+m^{2})}\right). \tag{9.2}$$

Here $\chi = gh$ and Eh are fixed dimensionless hardware and band parameters; the subscript on Θ allows its constants to depend on them. For m no larger than order $\sqrt{n}$, the n term sets the precision order. For m much larger than $\sqrt{n}$, the resolved-interaction term m^{2} dominates. A contact allowance proportional to n permits inverse-square risk. The corresponding root-mean-square error, the square root of the displayed risk, changes from inverse-square-root to inverse-linear order when that full temporal resource is supplied.

This improvement is attainable because a choice between plain and echoed cells produces records with a known, uniformly sensitive phase after the unknown direction cancels. Calibration and several interrogation scales resolve the phase's periodic ambiguity. The controller changes readout bases using previous estimates; it does not change the physical compensation. The converse follows a smooth nuisance curve and counts coherent derivative accumulation only where inside-cell access is supplied. Neither part is obtained by taking the reciprocal of a local Fisher information formula.

Timing brings a third distinction. A supplied known fixed displacement still specifies a pure conditional operation and a computable recorded phase. Hidden independent timing signs leave different operations mixed inside an observed parity branch. For the specified plain/one-pulse menu, fixed nonzero jitter makes the best global risk inverse-linear, even though a local quadratic response can survive at exact compensation. The necessary shrinking scale $d = O(h/\sqrt{n})$ identifies a restriction on quadratic global performance; it is not a construction proving that this accuracy suffices. Learning an unknown systematic displacement is also not included without a separate calibration experiment and its cost.

9.1 The assumptions are part of the answer

Several resources stay ideal throughout: instantaneous instruments, protected memory, a stable shared noise direction in the sensing task, and a known collision clock. The general discrimination theorem also assumes calibrated ν and Λ and the centered absence of coherent drift. Its simulator uses a potentially very large finite bath initialized at time zero and an externally prescribed switching schedule. The result does not establish efficient preparation of that bath or an autonomous implementation of its clock.

The two timing statements use different guarantees. The universal survival witness tolerates waiting intervals throughout a factor-two range tied to ν/Λ^{2}. The hidden-jitter sensing theorem concerns a particular unobserved two-point displacement of a particular pulse. A robust survival

test therefore does not imply that every global sensing protocol tolerates the same timing errors. Nor does the restricted jitter converse rule out better performance with extra pulses, observed timing labels, coupling selection, or other control resources outside its menu.

The short-time survival mechanism, channel dilations, collision models, adaptive channel bounds and multiscale phase estimation are established ingredients. The scientific content here is their use in the stated matching operational laws and fully charged sensing constructions, with the quantifiers and limits made explicit. It is not a claim that temporal resolution or dynamical decoupling was first recognized by this project.

9.2 What has been checked

The analytical proofs establish the universal statements. Finite calculations provide a different kind of evidence: they compare the exchange unitary with its reduced channel, test the exact echo, construct a noncommuting three-jump qutrit example, and propagate adaptive measurements while keeping every record. Separate sensing checks compare exact joint evolution with the phase formulas, the calibration likelihood, contact budgets and hidden timing mixtures. These checks are useful for finding sign, subsystem-order and bookkeeping errors; they do not optimize over every allowed controller.

The report's core message is therefore an operational one. A final decay curve describes what is visible at one endpoint. Temporal access determines which microscopic correlations can be tested while the dynamics runs and, in the specified sensor, which of those correlations can support more precise estimation. The answer depends on where the contacts occur, which timing information is retained, and which physical resources are actually supplied.